



# Chapter 4

## Differential Amplifiers

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# Outline

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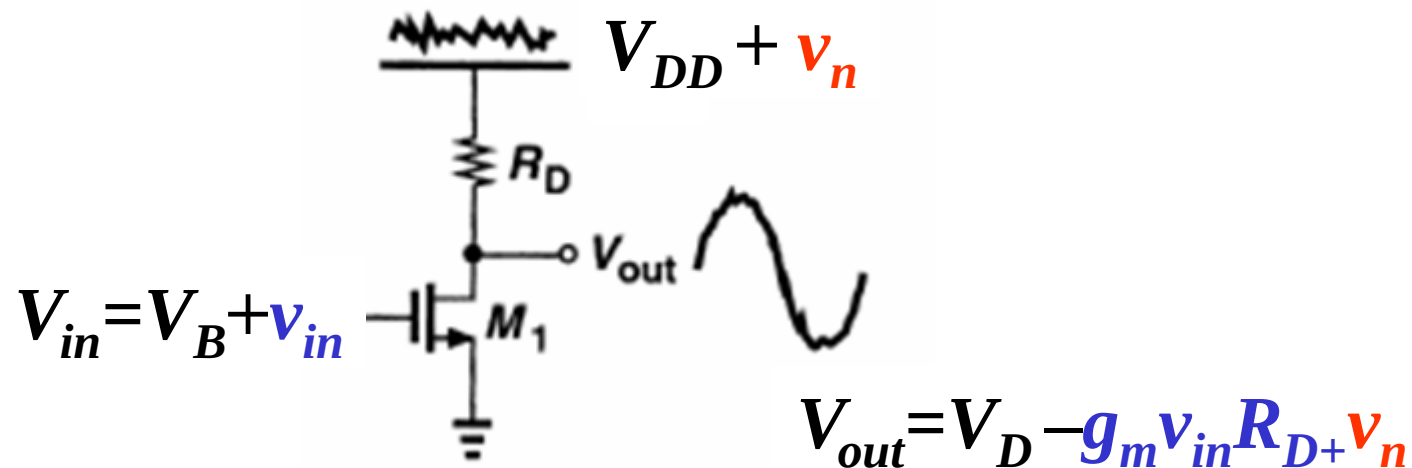
- **Single-Ended vs. Differential**
- Basic Differential Pair
- Common-Mode Response
- Differential Pair with MOS Loads
- Gilbert cell



# Analog circuit is sensitive to noise and interferences



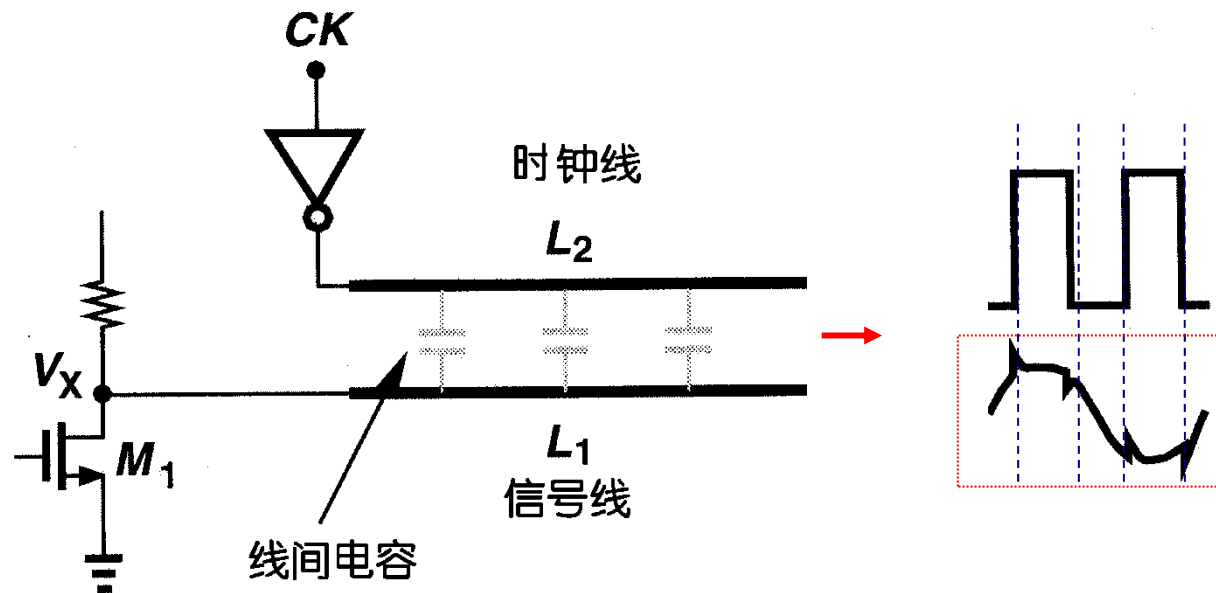
- Output signal is the sum of desired signal component and **unwanted noise**
  - MOS transistors generate noise
  - Resistors generate noise
  - $V_{DD}$  and GND are “noisy buses”



## Interference — an example: clock crosstalk



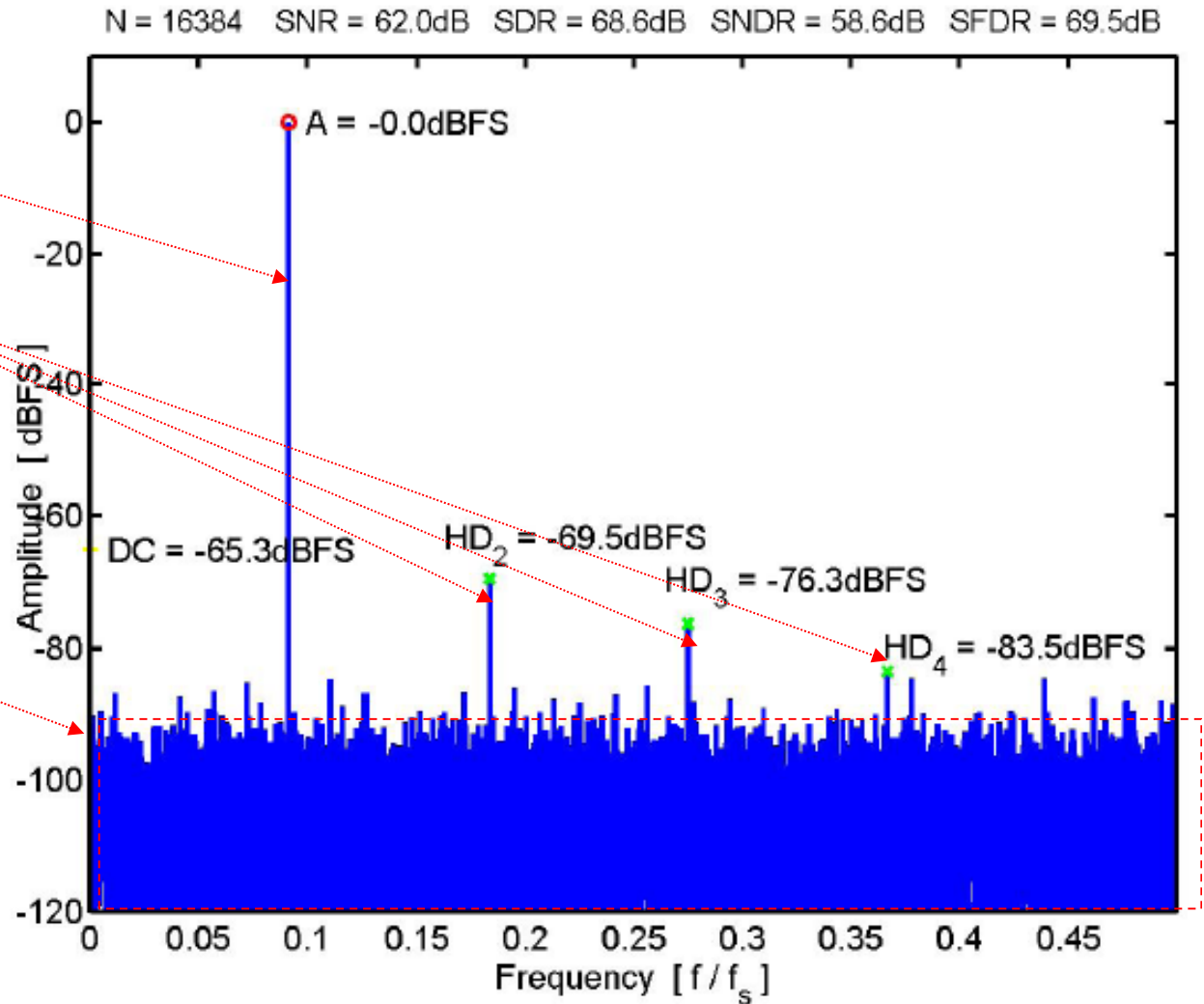
- Clock transitions are coupled to analog signals through parasitic capacitance
  - Desired signals are contaminated



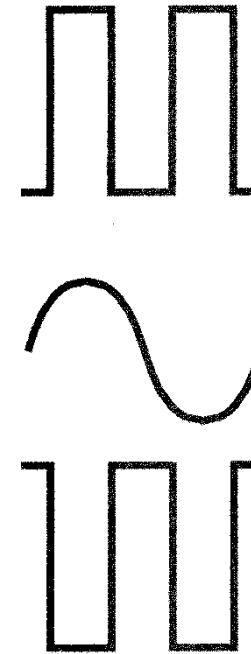
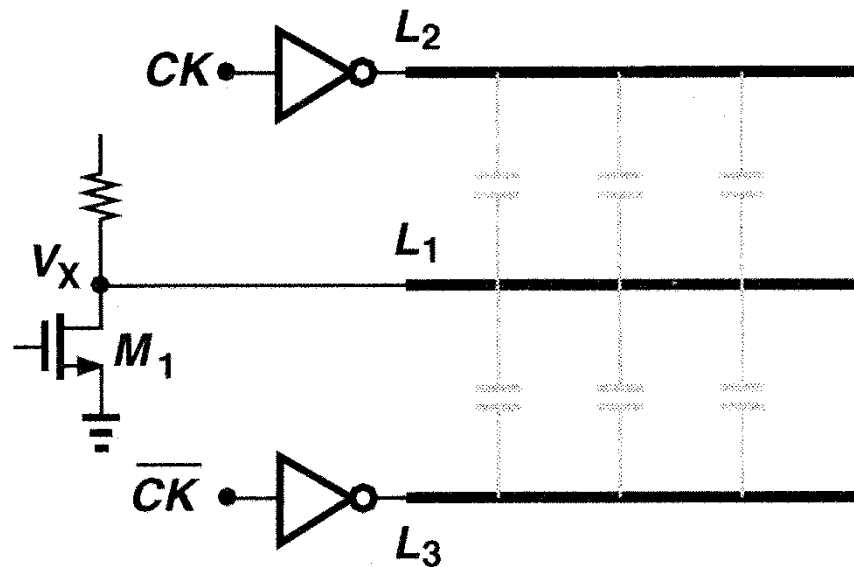
# Sinusoid signal **spectrum** from a analog system



- Signal
- Harmonics
  - nonlinearity
- Noise



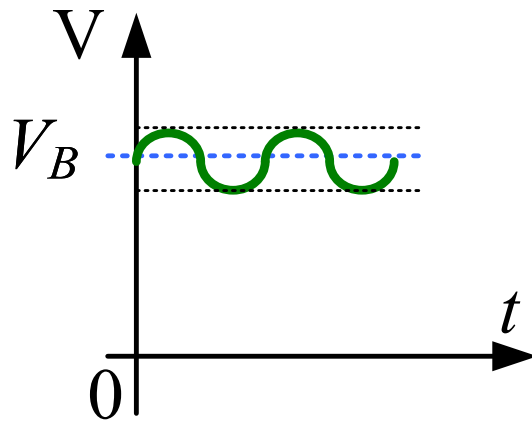
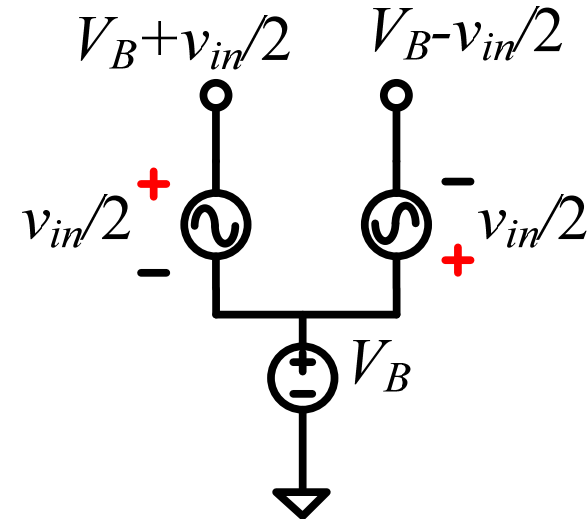
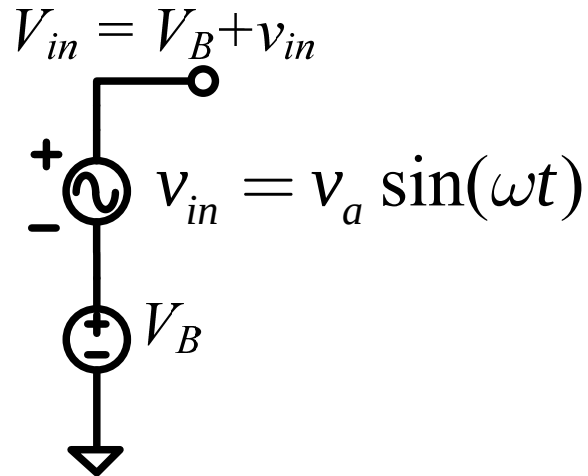
# Differential clock signals in digital signal



- The idea of differential (complementary) clock can also be employed in analog circuits



# Definition of differential signals



- **Common-mode level:**

$$V_{CM} = \frac{V_{in1} + V_{in2}}{2} = V_B$$

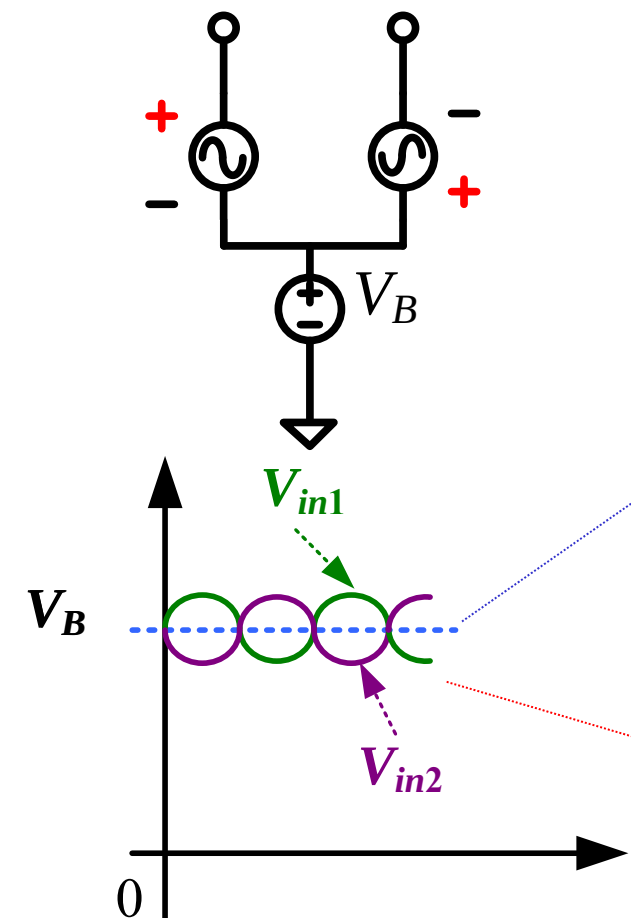
- **Differential-mode signal**

$$V_{diff} = V_{in1} - V_{in2} = v_{in}$$

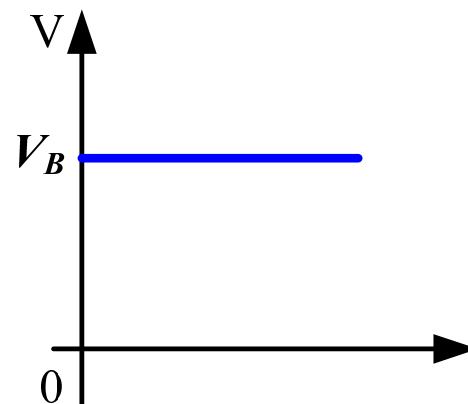




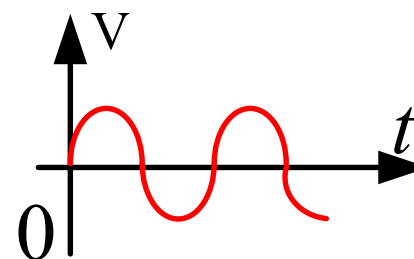
$$V_{in1} = V_B + v_{in} \quad V_{in2} = V_B - v_{in}$$



$$V_{CM} = \frac{V_{in1} + V_{in2}}{2} = V_B$$



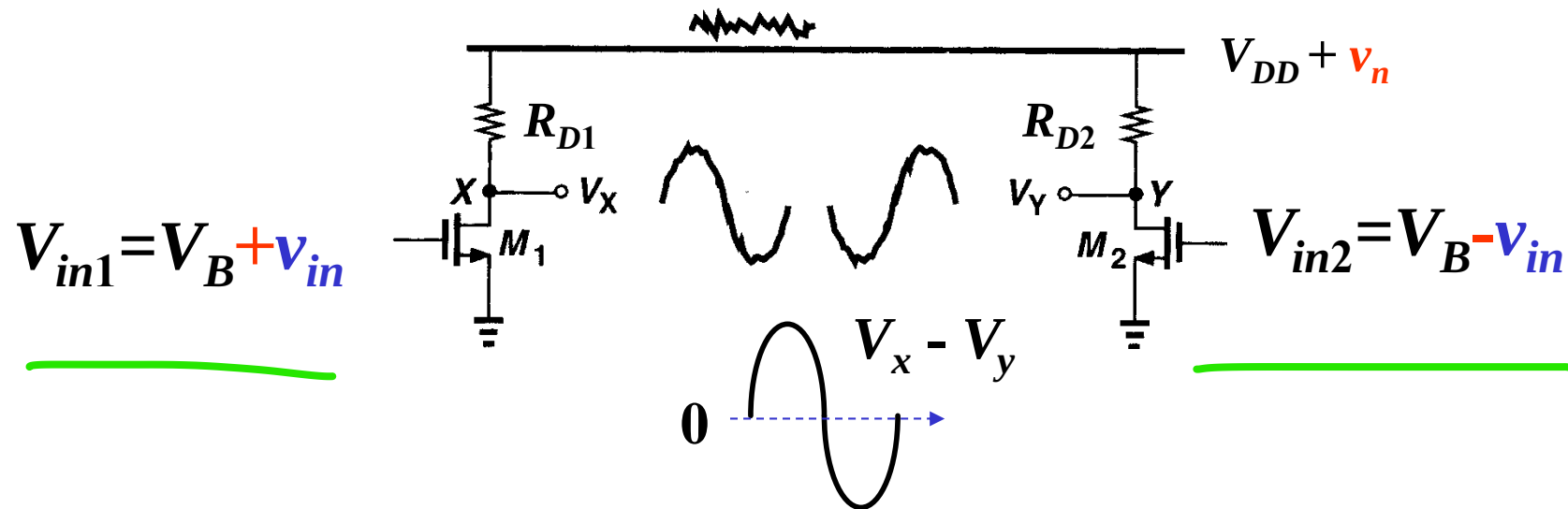
$$V_{diff} = V_{in1} - V_{in2} = 2v_{in}$$



# Noise canceling by differential signals



- Use “difference” of signals to cancel common-mode noise and interference



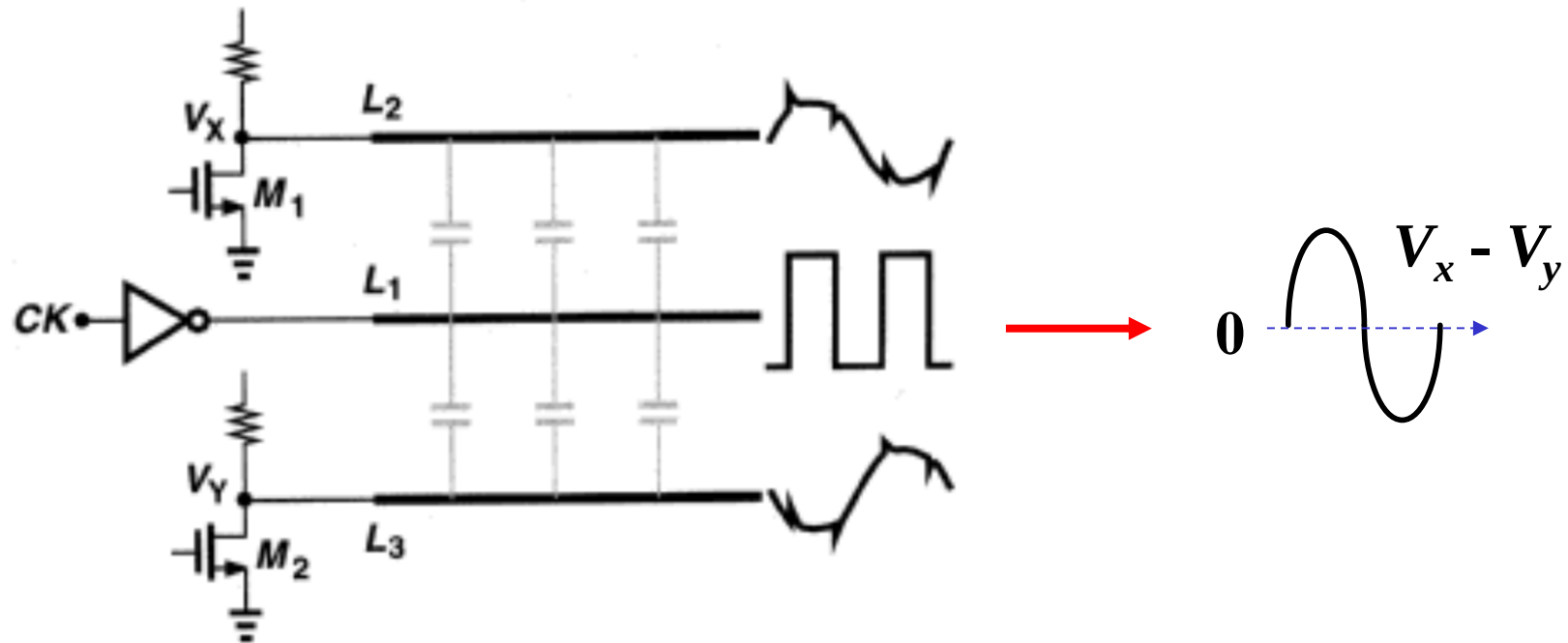
$$V_x = V_{D1} - g_{m1} v_{in} R_{D1} + v_n$$

$$R_{D1} = R_{D2} = R_D, \quad (W/L)_1 = (W/L)_2 \rightarrow g_{m1} = g_{m2} = g_m$$

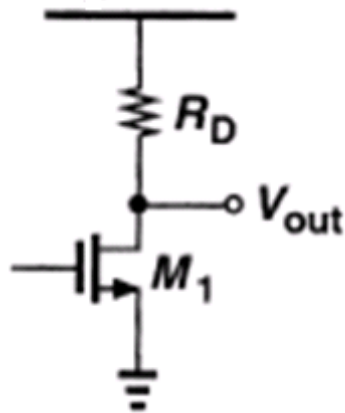
$$V_y = V_{D2} + g_{m2} v_{in} R_{D2} + v_n \rightarrow V_x - V_y = -2g_m v_{in} R_D$$



# Canceling of clock coupling with differential circuit



# Output swing of differential circuits

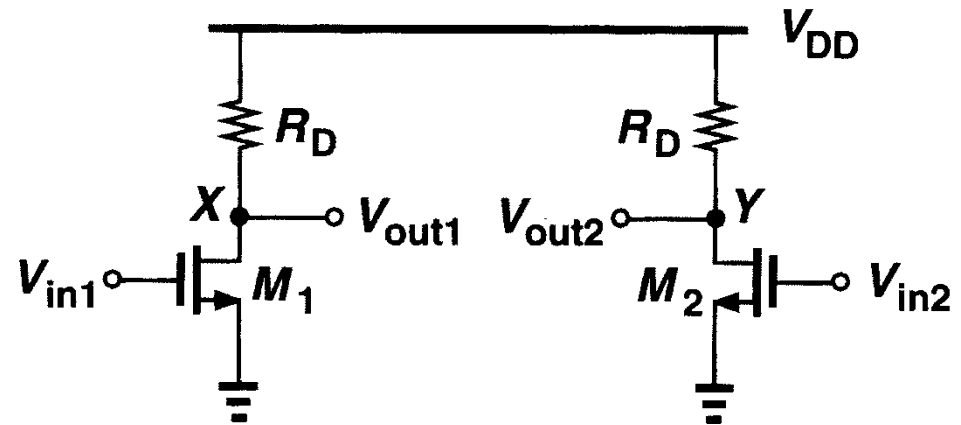


$$V_{out,max} = V_{DD}$$

$$V_{out,min} = V_{GS} - V_{TH}$$

**Swing of  $V_{out}$**

$$V_{DD} - V_{GS} + V_{TH}$$



$$V_{out1,max} = V_{DD}$$

$$V_{out2,min} = V_{GS} - V_{TH}$$

$$V_{out1,min} = V_{GS} - V_{TH} \quad V_{out2,max} = V_{DD}$$

**Swing of  $V_{out1} - V_{out2}$**

$$2(V_{DD} - V_{GS} + V_{TH})$$



# Summary: features of differential circuits

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- **Advantages:**

- Common-mode noise rejection
- Common-mode interference rejection
- Output swing is enlarged
- .....

- **Disadvantages:**

- Chip area and power are nearly doubled
- *Noise contribution increases*
- However, numerous advantages outweigh the increase in area and power dissipation

***Differential amplifier is one of the most important circuit inventions!!***



# Outline

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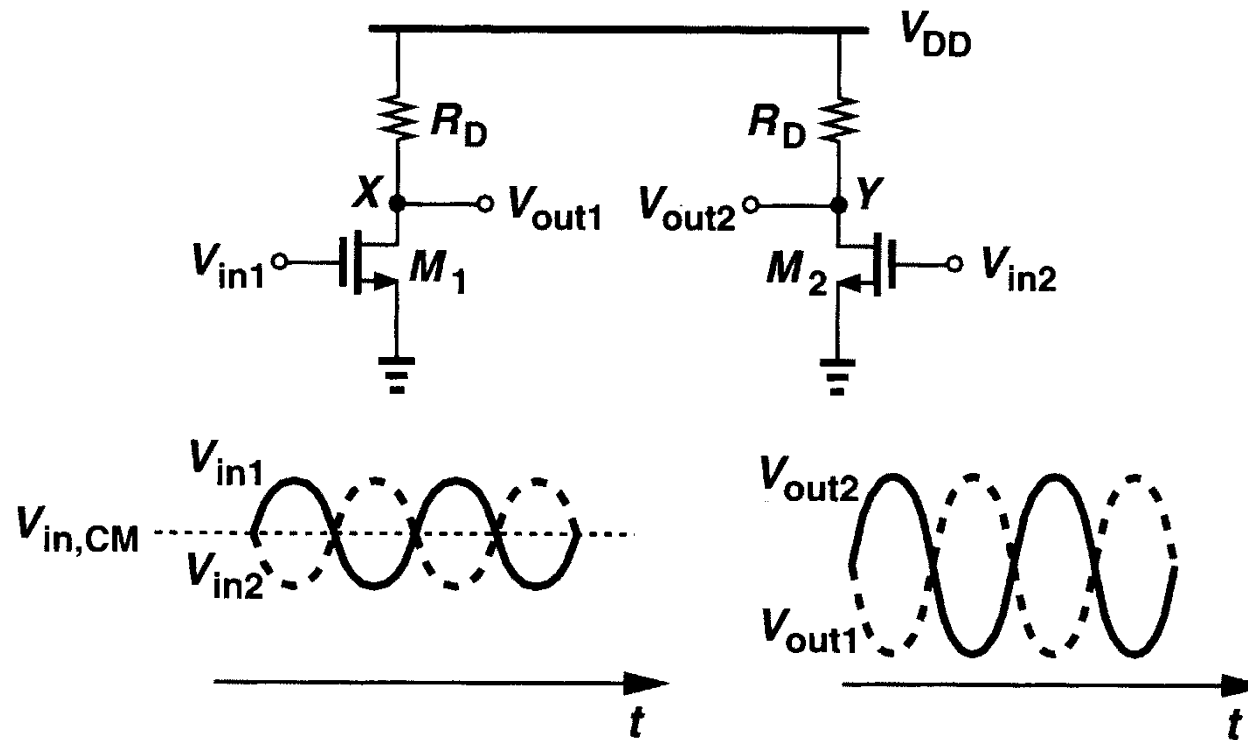


- Single-Ended vs. Differential
- **Basic Differential Pair**
- Common-Mode Response
- Differential Pair with MOS Loads
- Gilbert Cell

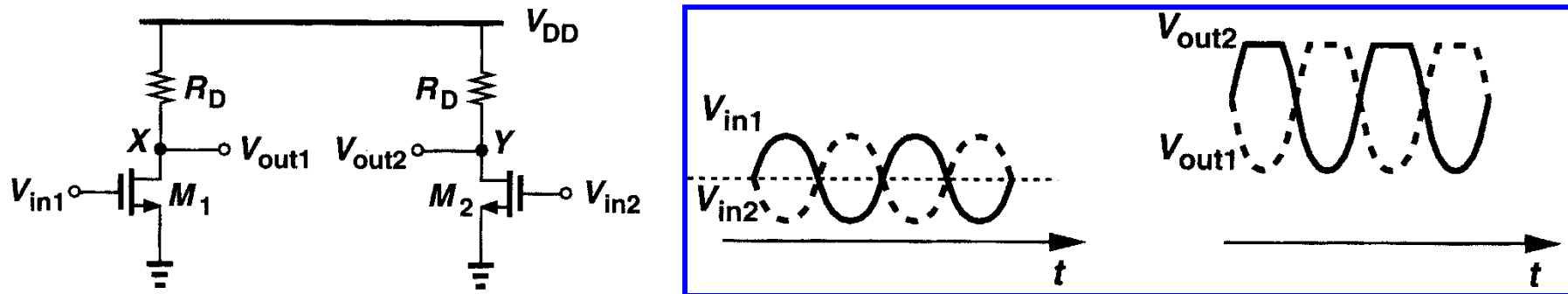
# Basic differential amplifier



- Two identical single-ended amplifiers
  - One processes the  $0^\circ$  input signal, the other processes the  $180^\circ$  signal



# Problem of the basic differential amplifier



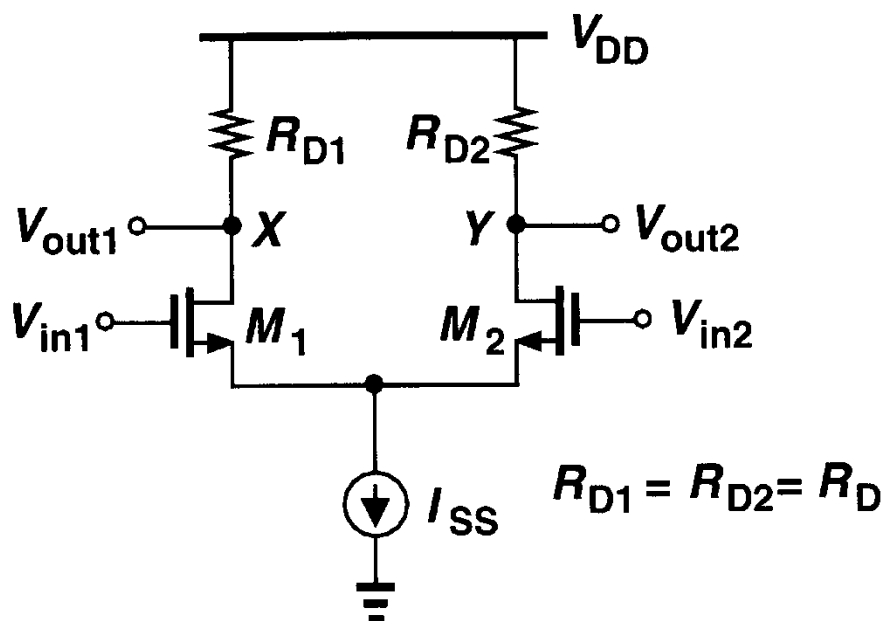
- Sensitive to variations in common-mode level
  - The operation point is determined by the common-mode level
  - Output signal may be “clipped”
  - Often called “pseudo differential”
- Solution: employ tail current source



## Differential amplifier with tail current source



- DC currents in M1 and M2 are determined by  $I_{SS}$ , independent of input common-mode level



If  $V_{in1} = V_{in2}$ ,

$$I_{D1} = I_{D2} = I_{SS} / 2$$

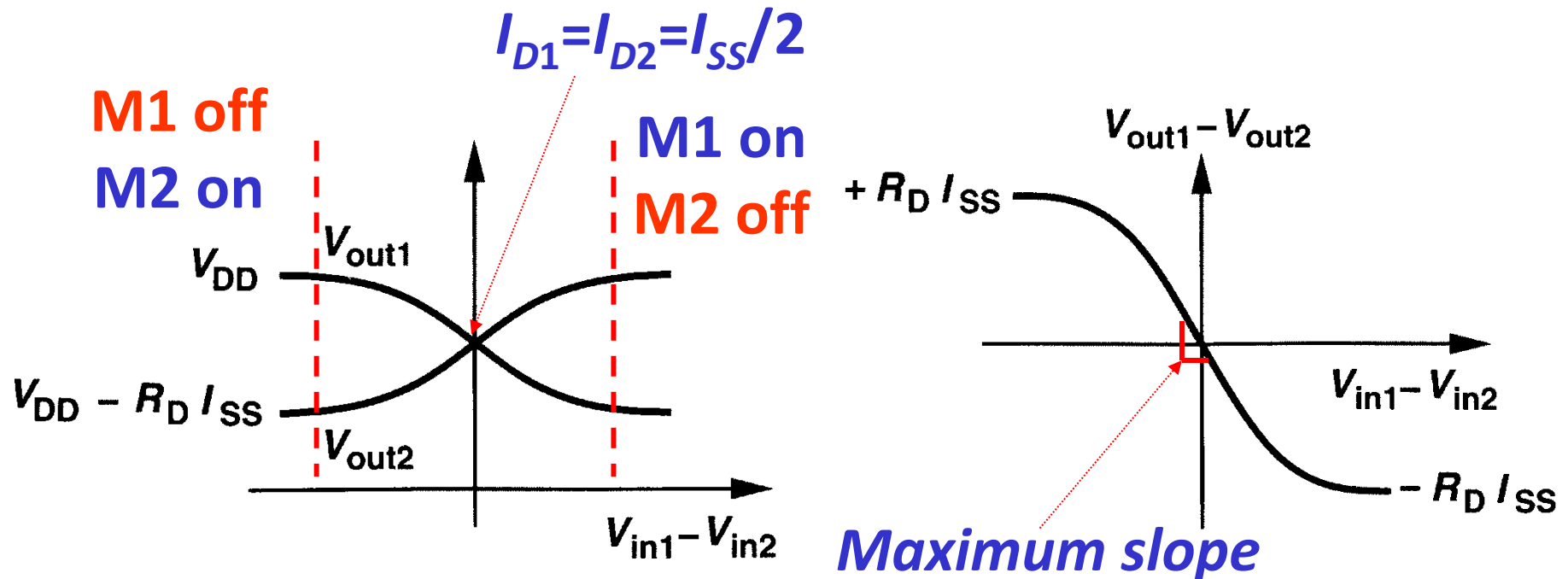
Output CM level is

$$V_{out,CM} = V_{DD} - R_D I_{SS} / 2$$

## Qualitative analysis



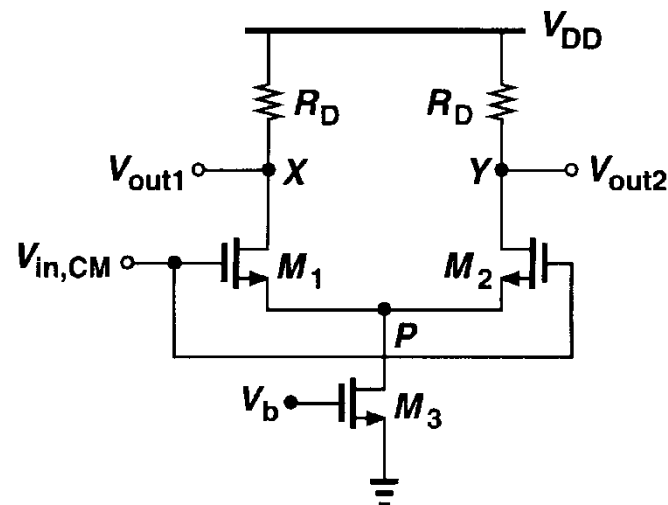
- important: consider all the signals differentially!
  - If  $V_{in1}-V_{in2}$  varies from  $-\infty$  to  $+\infty$
  - The  $I_{SS}$  is commutated from one to the other



## Common-mode behavior (1)



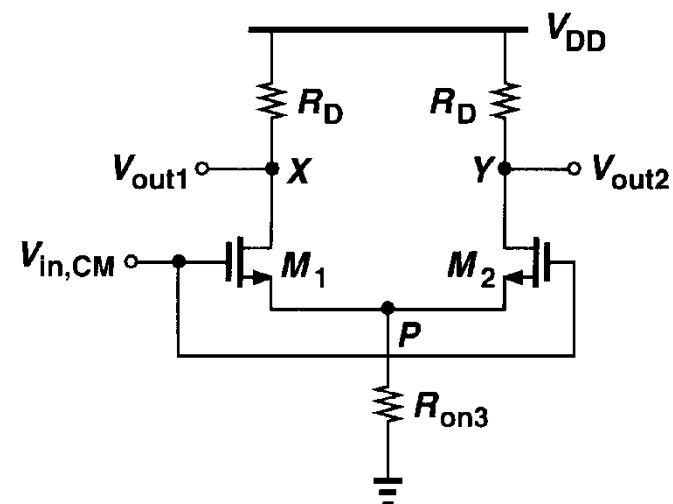
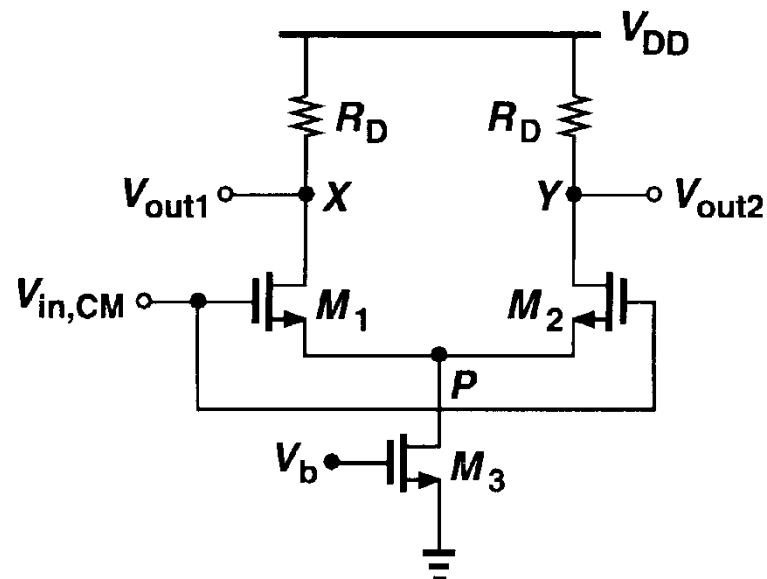
- Is the differential pair really independent of  $V_{in,CM}$ ?
  - If  $V_{in,CM}$  varies from 0 to  $V_{DD}$ , what happens with  $V_P$ ,  $V_{out1}$ ,  $V_{out2}$ , and  $I_{D1}$ ,  $I_{D2}$ ?
  - When  $V_{in,CM}=0$ :
    - $M_1$  and  $M_2$  are off,  $I_{D1,2}=0$ ,  $V_{out1,2}=V_{DD}$ ,  $V_P=0$



## Common-mode behavior (2)



- When  $V_{in,CM} \geq V_{TH}$ 
  - $M_1$  and  $M_2$  are on,  $M_3$  operates in **triode region** and can be **regarded as a resistor**



$V_{in,CM}$  is small  
 $M_3$  operates in triode region

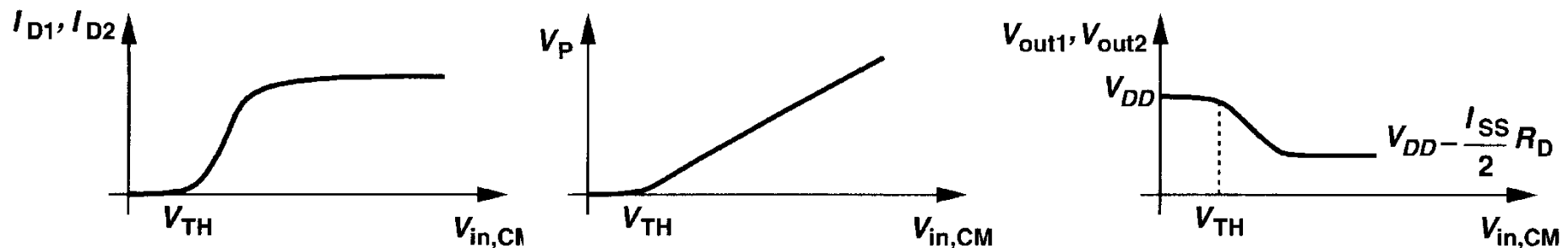
## Common-mode behavior (3)



- To ensure proper operation:

$$V_{in,CM} \geq V_{GS1} + (V_{GS3} - V_{TH3})$$

- If  $V_{in,CM}$  rises further
  - $I_{D1,2}$  and  $V_{out1,2}$  are relatively constant
  - $M_{1,2}$  and  $M_3$  operates as a source follower



## Common-mode behavior (4)



- To ensure  $M_{1,2}$  operate in saturation

$$V_{in,CM} \leq V_{out1,2} + V_{TH} = V_{DD} - R_D I_{SS}/2 + V_{TH}$$

- The operation range of  $V_{in,CM}$  is concluded as:

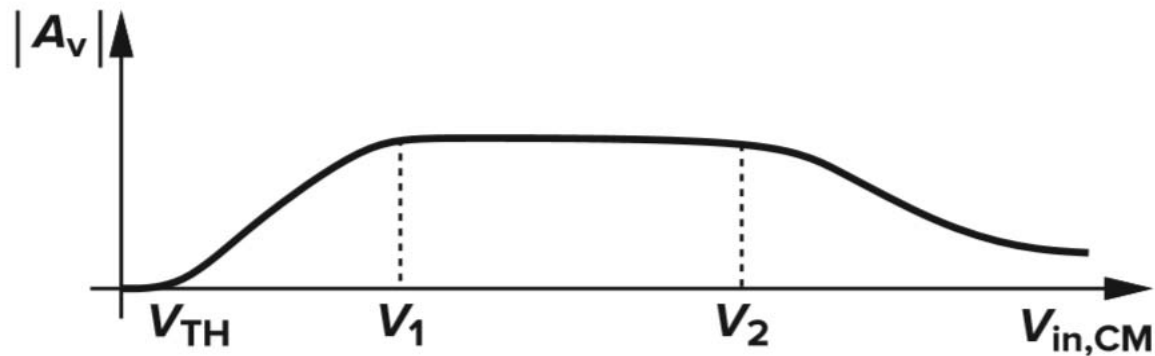
$$V_{GS1} + V_{GS3} - V_{TH3} \leq V_{in,CM} \leq \min[V_{DD}, V_{DD} - R_D I_{SS}/2 + V_{TH}]$$



## Ex. 4.2



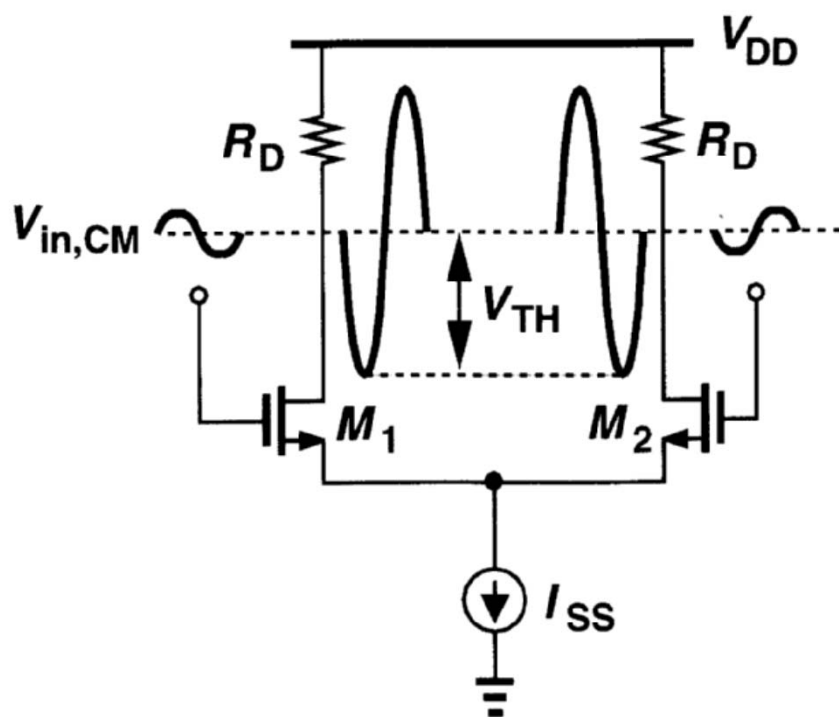
- Sketch the differential small-signal gain as a function of  $V_{in,CM}$



- $V_{in,CM} < V_1$ , tail current source is in linear region
- $V_{in,CM} > V_2$ , input pair is in linear region



# Common-Mode Input vs. Output Swing



Output swing: ( $V_{in,CM}$  is constant.)

Max:  $V_{DD}$

Min:  $\max(V_{in,CM} - V_{TH}, V_{DD} - I_{SS} R_D)$

Input CM voltage swing :

( $R_D$  is constant.)

Max:  $\min(V_{DD}, V_{DD} - \frac{1}{2} I_{SS} R_D + V_{TH})$

Min:  $V_{ISS} + V_{GS1,2}$

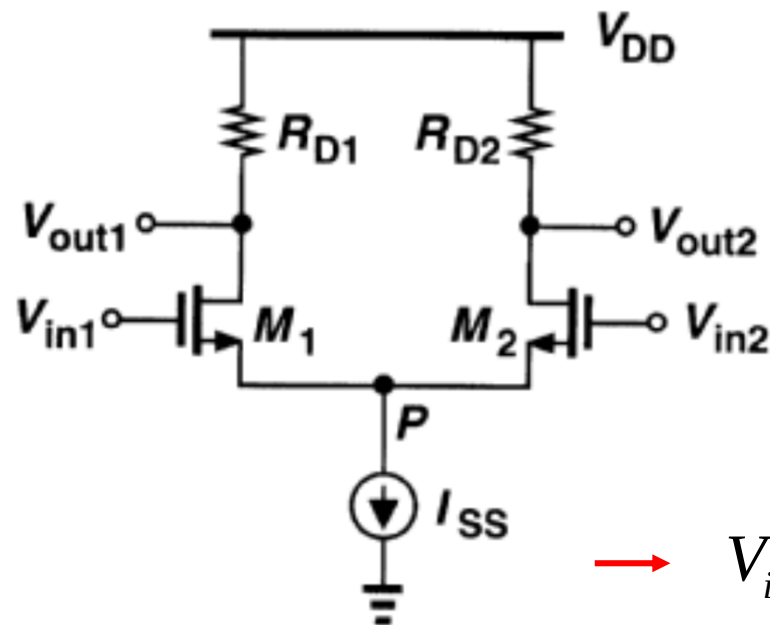


# Quantitative analysis: large-signal behavior



$$\begin{aligned} V_{out1} - V_{out2} &= (V_{DD} - I_{d1}R_{D1}) - (V_{DD} - I_{d2}R_{D2}) \\ &= -(I_{d1} - I_{d2})R_D \end{aligned}$$

- Derive the relationship between  $V_{in1} - V_{in2}$  and  $I_{d1} - I_{d2}$



$$V_{in1} - V_{gs1} = V_{in2} - V_{gs2} = V_P$$

$$\rightarrow V_{in1} - V_{in2} = V_{gs1} - V_{gs2}$$

$$\therefore V_{gs} = \sqrt{\frac{2I_d}{\mu_n C_{ox} W/L}} + V_{TH}$$

$$\rightarrow V_{in1} - V_{in2} = \sqrt{\frac{2I_{d1}}{\mu_n C_{ox} W/L}} - \sqrt{\frac{2I_{d2}}{\mu_n C_{ox} W/L}}$$



$$(V_{in1} - V_{in2})^2 = \frac{2}{\mu_n C_{ox} W / L} (I_{SS} - 2\sqrt{I_{d1} I_{d2}})$$

$$\therefore 4I_{d1} I_{d2} = (I_{d1} + I_{d2})^2 - (I_{d1} - I_{d2})^2$$

$$\rightarrow I_{d1} - I_{d2} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{in2}) \sqrt{\frac{4I_{SS}}{\mu_n C_{ox} W / L} - (V_{in1} - V_{in2})^2}$$

- Define  $I_{d1} - I_{d2} = I_{d,diff}$ ,  $V_{in1} - V_{in2} = V_{in,diff}$   
– Equivalent  $G_m$  is calculated as

$$G_m = \frac{\partial I_{d,diff}}{\partial V_{in,diff}} \rightarrow$$

Control ability of differential input voltage on differential output current





$$G_m = \frac{\partial I_{d,diff}}{\partial V_{in,diff}} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \frac{\frac{4I_{SS}}{\mu_n C_{ox} W / L} - 2V_{in,diff}^2}{\sqrt{\frac{4I_{SS}}{\mu_n C_{ox} W / L} - V_{in,diff}^2}}$$

- When  $V_{in,diff} = 0$

→  $G_m = \sqrt{I_{SS} \mu_n C_{ox} \frac{W}{L}} = \sqrt{2 \frac{I_{SS}}{2} \mu_n C_{ox} \frac{W}{L}} = g_{m1} = g_{m2}$

- Differential small-signal gain is expressed as :

$$A_{v,diff} = -G_m R_D = -\sqrt{I_{SS} \mu_n C_{ox} W / L} \cdot R_D$$



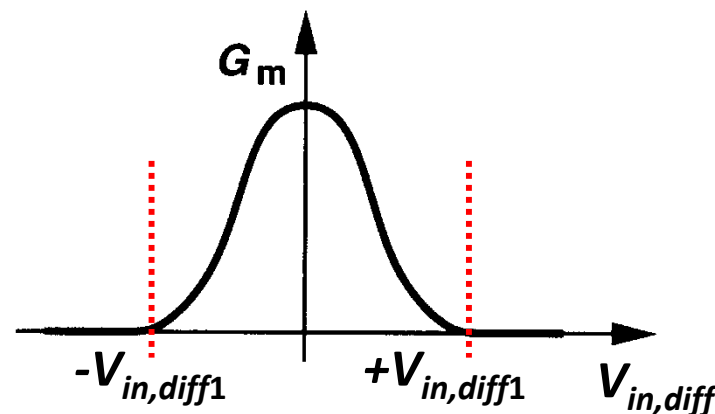


## $G_m$ as a function of $V_{in,diff}$

$$G_m = \frac{\partial I_{d,diff}}{\partial V_{in,diff}} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \frac{\frac{4I_{SS}}{\mu_n C_{ox} W / L} - 2V_{in,diff}^2}{\sqrt{\frac{4I_{SS}}{\mu_n C_{ox} W / L} - V_{in,diff}^2}}$$

- $G_m$  is at maximum when  $V_{in,diff} = 0$ 
  - $G_m$  falls to 0 when  $V_{in,diff} = V_{in,diff1}$

$$|V_{in,diff1}| = \sqrt{\frac{2I_{SS}}{\mu_n C_{ox} W / L}}$$



## Why $G_m$ falls to 0 when $V_{in,diff} = V_{in,diff1}$ ?



- Finding the currents at  $V_{in,diff1}$ , recall that

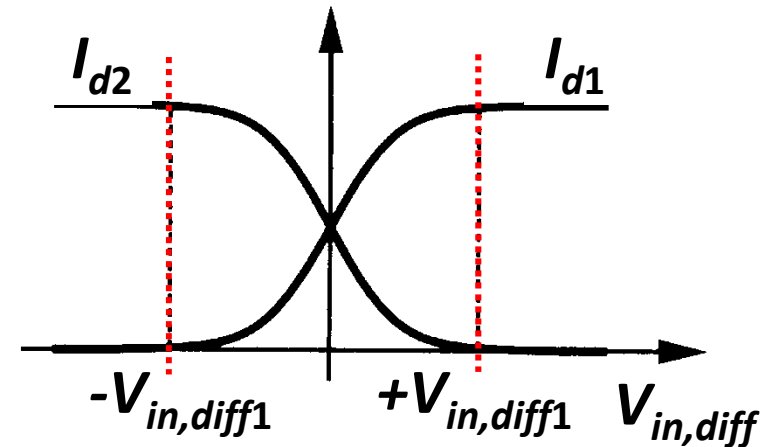
$$|V_{in,diff1}| = |V_{in1} - V_{in2}| = \left| \sqrt{\frac{2I_{d1}}{\mu_n C_{ox} W / L}} - \sqrt{\frac{2I_{d2}}{\mu_n C_{ox} W / L}} \right| = \sqrt{\frac{2I_{SS}}{\mu_n C_{ox} W / L}}$$

$$I_{d1} + I_{d2} = I_{SS}$$

- The only valid solution is either

$$I_{d1} = I_{SS}, I_{d2} = 0 \quad \text{or} \quad I_{d1} = 0, I_{d2} = I_{SS}$$

**Why?**



- The tail current flows through only one transistor while the other is off, so control ability is lost



## Valid differential input range



- In order not to lose control ability

$$|V_{in,diff}| < \sqrt{\frac{2I_{SS}}{\mu_n C_{ox} W / L}} = \sqrt{2} \sqrt{\frac{2(I_{SS} / 2)}{\mu_n C_{ox} W / L}}$$

$$|V_{in,diff}| < \sqrt{2} \cdot (V_{GS} - V_{TH})_{at V_{in,diff}=0}$$

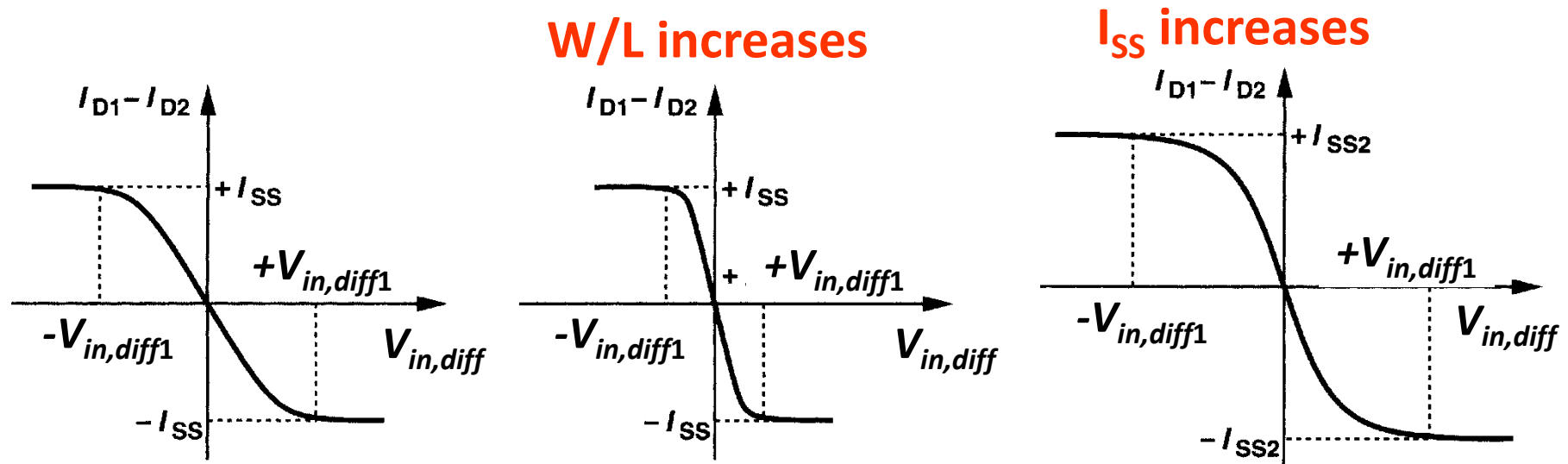
- The input range is determined by  $(V_{GS} - V_{TH})_{at V_{in,diff}=0}$
- What is this voltage dependent on?
  - Tail current and input transistor size



## Example 4.2



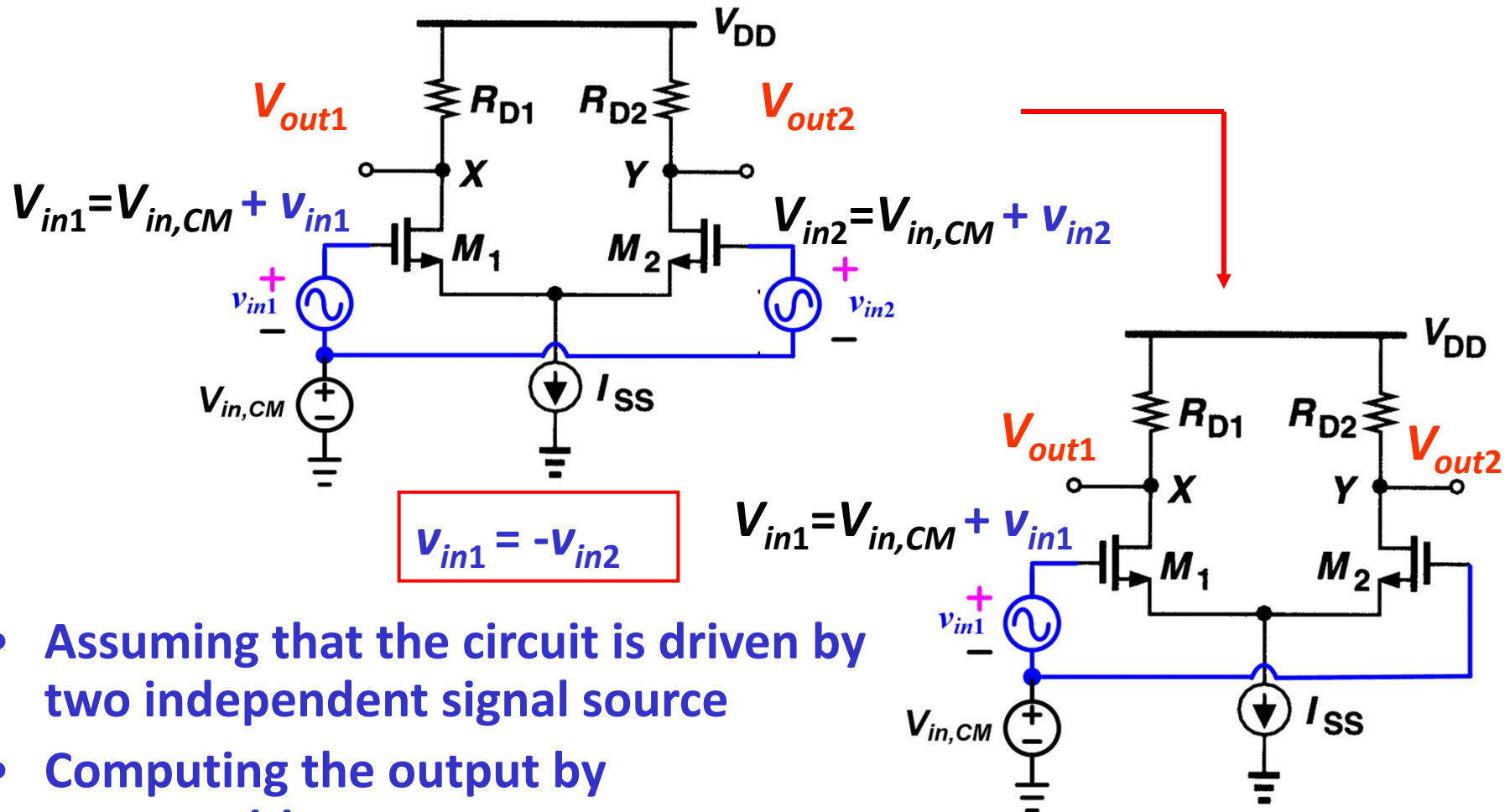
- Influences of device size and tail current on the input/output characteristics



- The input range can be enlarged by either reducing W/L or increasing  $I_{SS}$



# Small-signal gain method I: superposition



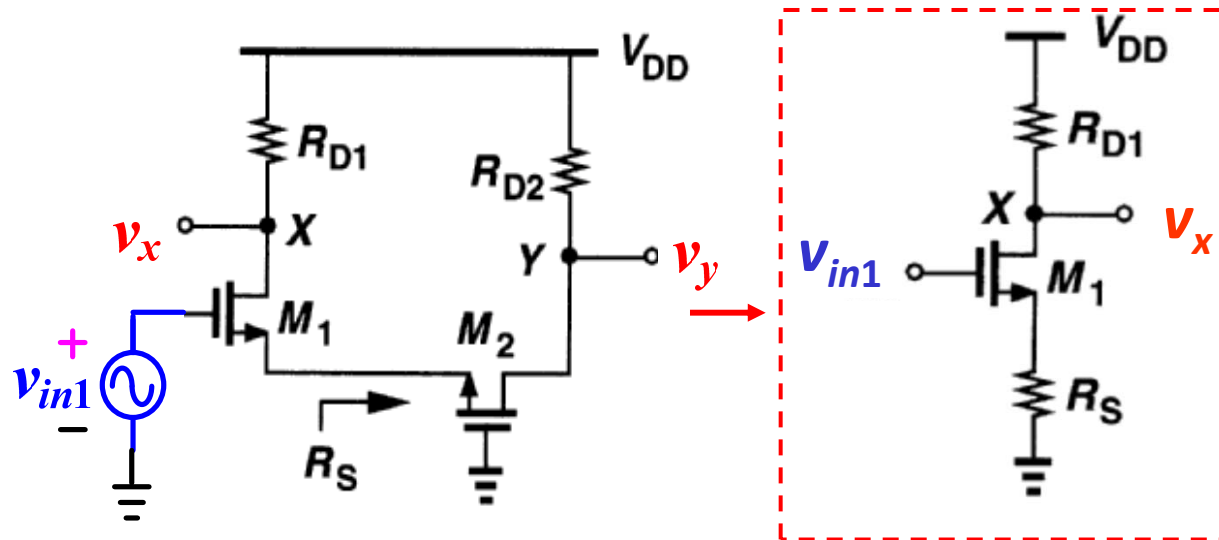
- Assuming that the circuit is driven by two independent signal source
- Computing the output by superposition



## Superposition method (2)



- Consider only small signal behavior
  - Calculating  $v_x$  due to  $v_{in1}$



Assuming  $\lambda = \gamma = 0$

$$R_S \approx 1 / g_{m2}$$

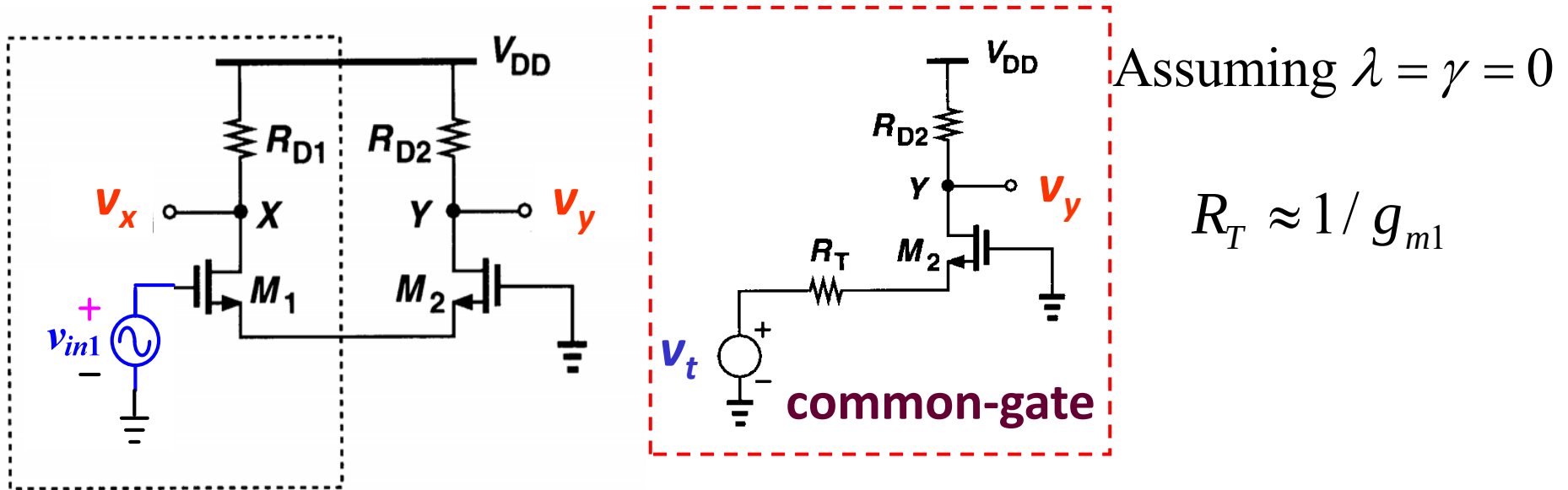
$$\frac{v_x}{v_{in1}} = -\frac{g_{m1}}{1 + g_{m1}R_S} R_D = -\frac{R_D}{1 / g_{m1} + 1 / g_{m2}}$$



## Superposition method (3)



- Calculating  $v_y$  due to  $v_{in1}$ 
  - M1 drives M2 as a source follower
  - Replace it by a Thevenin equivalent



$$\frac{v_y}{v_{in1}} = \frac{g_{m2}}{1 + g_{m2} R_T} R_D = \frac{R_D}{1 / g_{m1} + 1 / g_{m2}}$$



## Superposition method (4)



- **Computing the output by superposition**

- **Output due to  $v_{in1}$**

$$v_x = \frac{-R_D v_{in1}}{1/g_{m1} + 1/g_{m2}}, \quad v_y = \frac{R_D v_{in1}}{1/g_{m1} + 1/g_{m2}}$$

$$(v_x - v_y) \big|_{\text{due to } v_{in1}} = \frac{-2R_D}{1/g_{m1} + 1/g_{m2}} \cdot v_{in1} = -g_m R_D \cdot v_{in1}$$

- **Similarly, output due to  $v_{in2}$**

$$(v_x - v_y) \big|_{\text{due to } v_{in2}} = \frac{2R_D}{1/g_{m1} + 1/g_{m2}} \cdot v_{in2} = g_m R_D \cdot v_{in2}$$

$$\rightarrow A_v = \frac{(v_x - v_y) \big|_{\text{tot}}}{v_{in1} - v_{in2}} = -g_m R_D$$

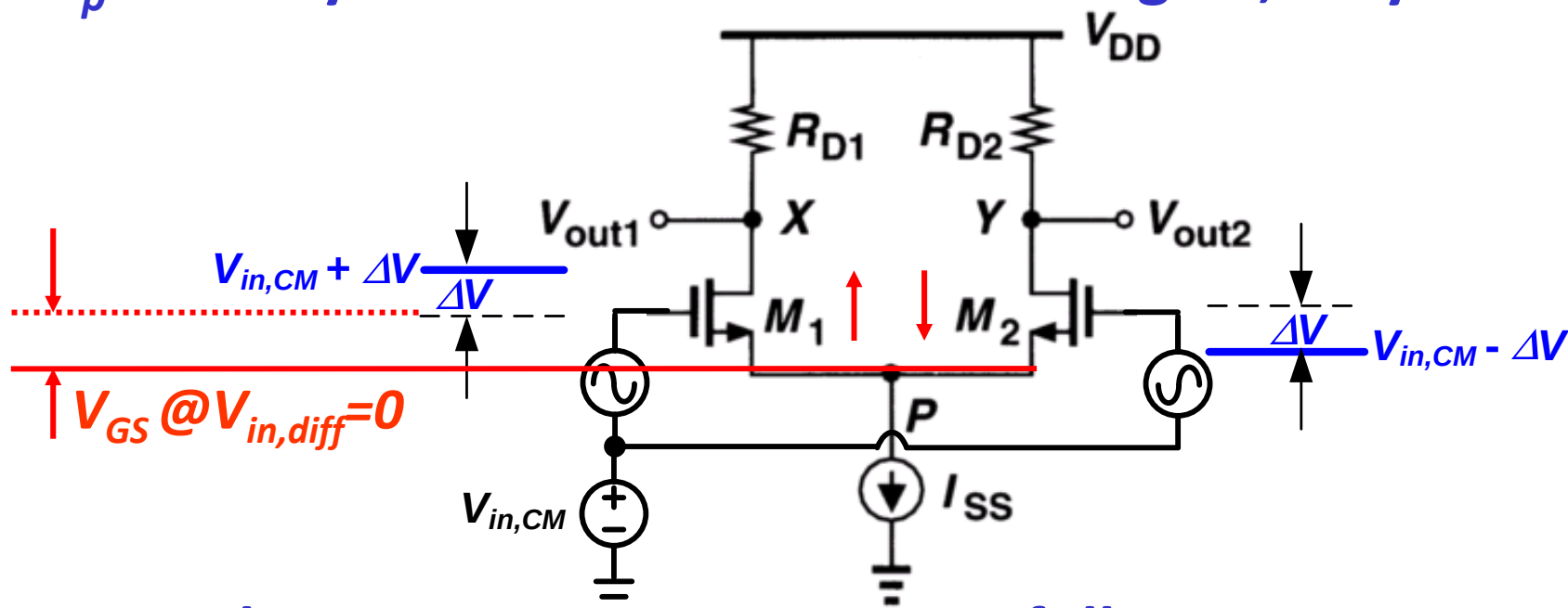
**The signal can be applied either at one side or differentially!**



## Small-signal gain-method 2: half circuit



$V_p$  is always constant for differential signal, why?

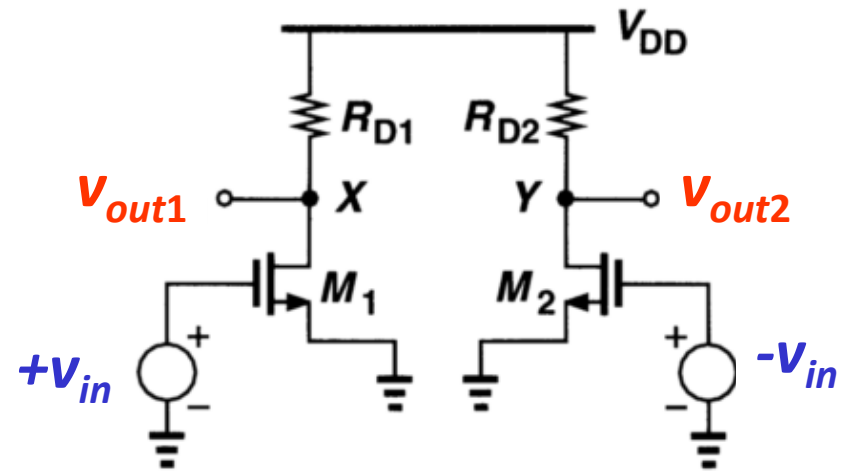
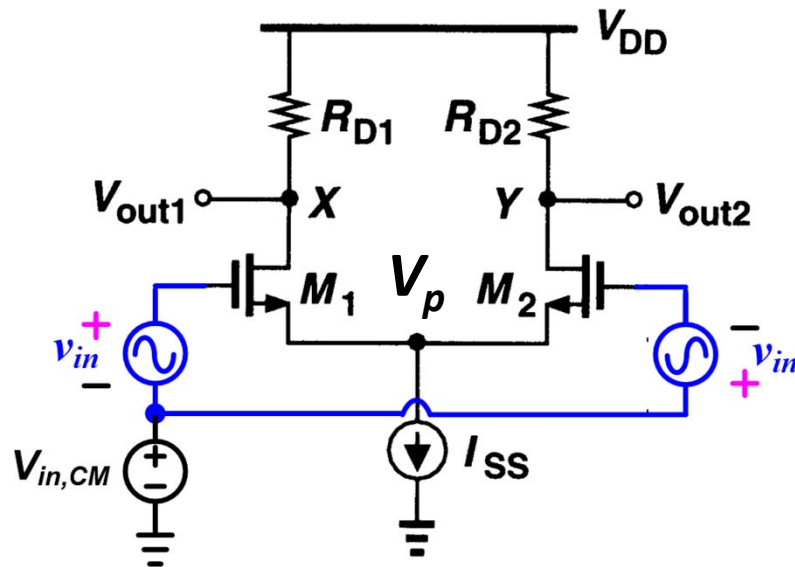


- $M_1$  and  $M_2$  operate as two source followers
  - $V_{in1}$  increases by  $\Delta V \rightarrow V_p$  increases by  $\Delta V$
  - $V_{in2}$  decreases by  $\Delta V \rightarrow V_p$  decreases by  $\Delta V$
- $V_p$  keeps constant if  $M_1$  and  $M_2$  are fully symmetric

## Half circuit (2)



Because  $V_p$  is assumed to be constant for differential large-signal, it can be regarded as a small-signal ground



$$v_{out1}/v_{in} = -g_m R_D$$

$$v_{out2}/(-v_{in}) = -g_m R_D$$

$$A_v = \frac{v_{out1} - v_{out2}}{2v_{in}} = -g_m R_D$$

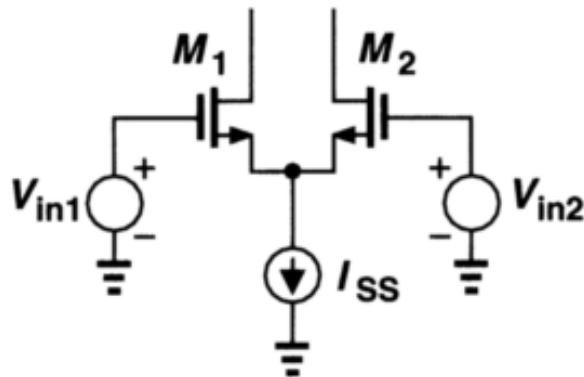
Effective analysis  
method for diff. circuit



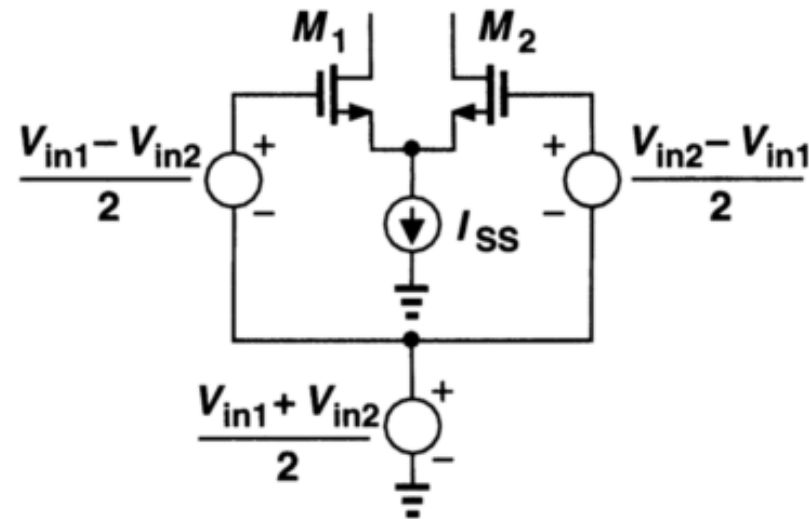
## Decomposing arbitrary input signals of diff. pair



- For arbitrary input signals to a differential pair
  - Can be decomposed into common-mode level  $V_{in,CM}$  and differential-mode signal  $V_{in,DM}$



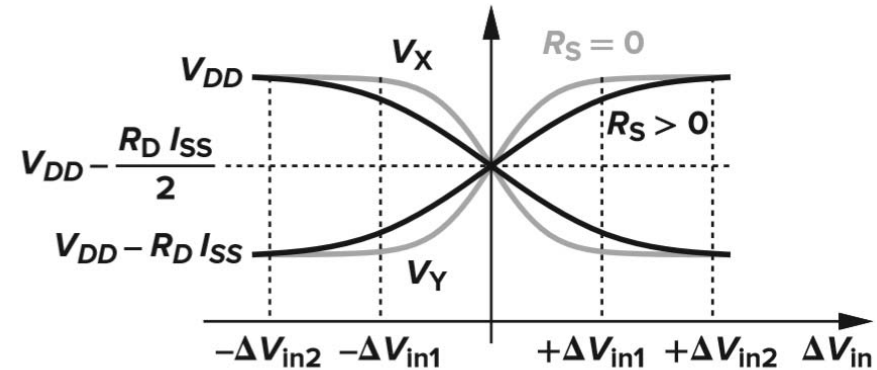
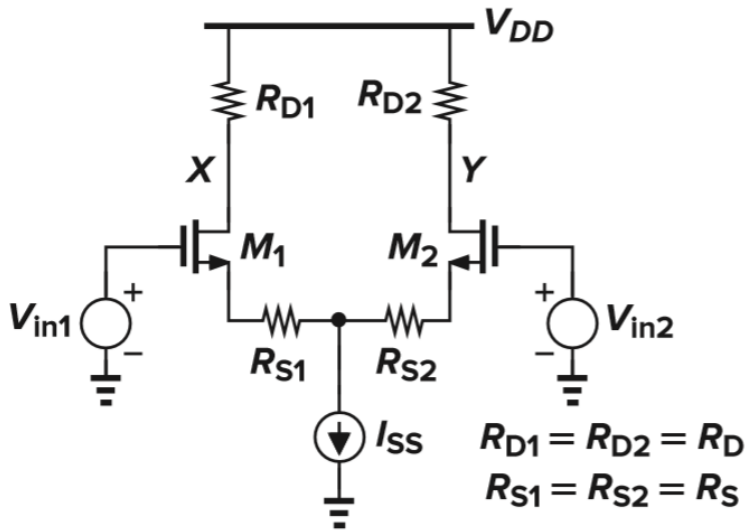
$$V_{in,CM} = \frac{V_{in1} + V_{in2}}{2}$$



$$V_{in,DM} = V_{in1} - V_{in2}$$



# Differential pair with source degeneration



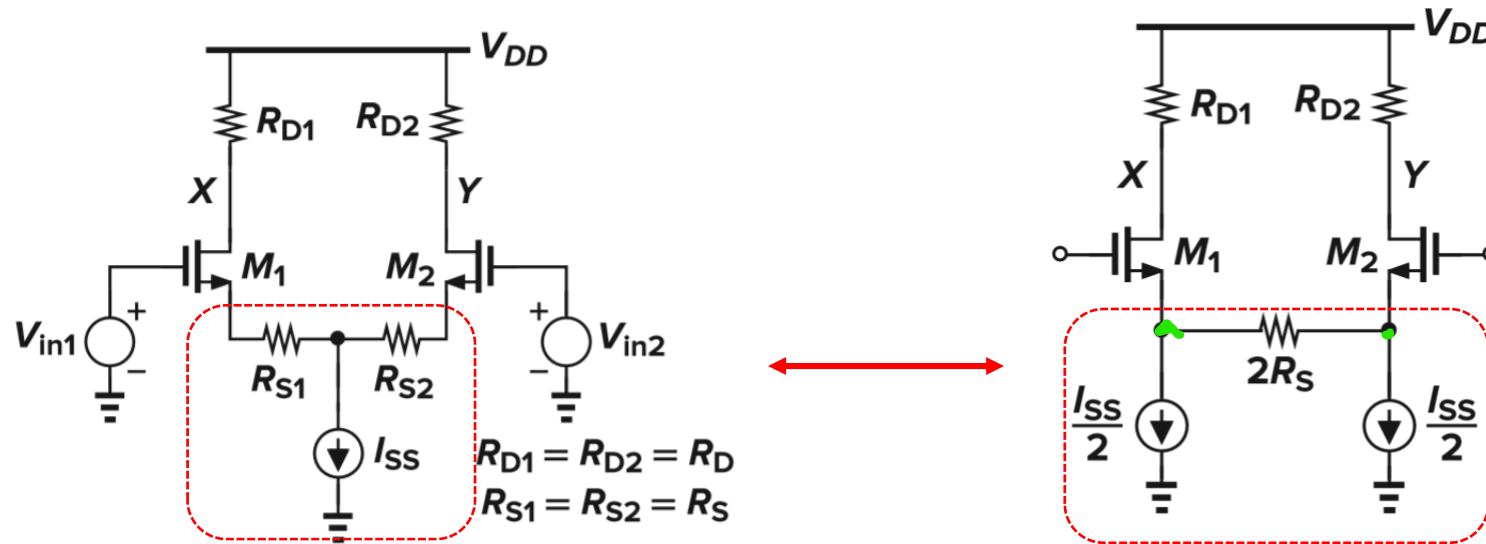
$$V_{in1} - V_{in2} = V_{GS1} - V_{TH} + R_S I_{SS}$$

- Improve the linearity
  - At the price of lower gain
  - Increased input range, assuming  $V_{in1} > V_{in2}$ , and  $I_{SS} = I_{D1}$

$$= \sqrt{\frac{2I_{SS}}{\mu_n C_{ox} \frac{W}{L}}} + \boxed{R_S I_{SS}}$$



# Source-degenerated diff. pair without DC loss



Output swing loss because of the DC drop on  $R_S$

- No DC drop on  $R_S$
- Noise contribution increases

$$A_v = \frac{g_{m1}}{1 + g_{m1} R_S} R_D$$



# Outline

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- Single-Ended vs. Differential
- Basic Differential Pair
- **Common-Mode Response**
- Differential Pair with MOS Loads
- Gilbert cell



## Common-Mode Response: ideal case

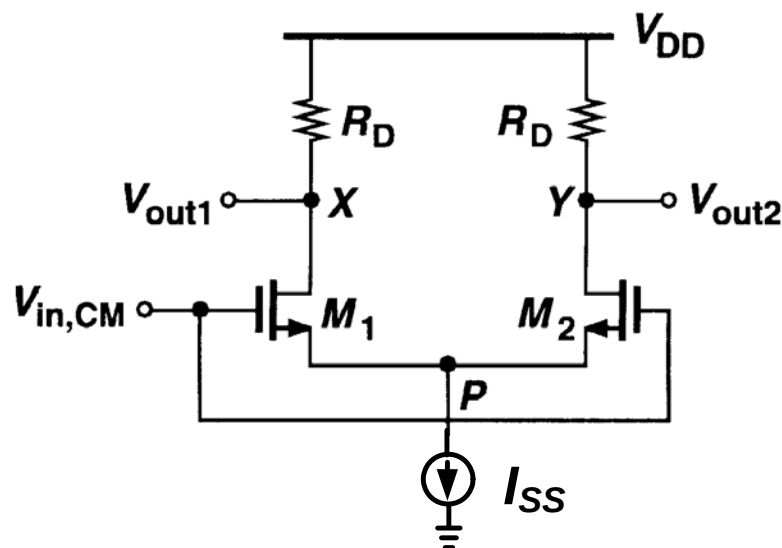
- The response of  $V_{out,CM}$  to  $V_{in,CM}$  is a very important performance of differential pairs

– *Ideal case:*

- Fully symmetrical ; ideal tail current source

- If  $V_{in,CM}$  satisfies  $V_{in,CM} \geq V_{GS1}$

– The current in  $R_D$  is independent of  $V_{in,CM}$



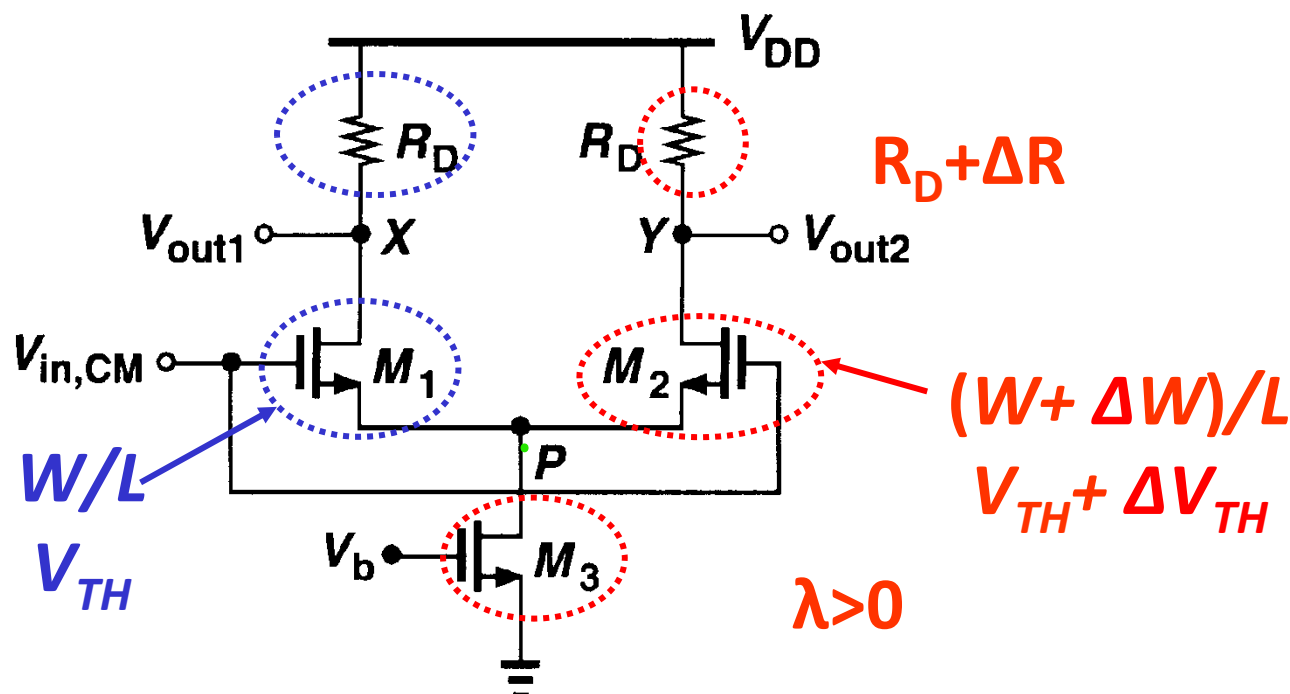
$$V_{out,CM} = \frac{V_{out1} + V_{out2}}{2}$$
$$= V_{DD} - I_{SS} R_D / 2 = \text{constant}$$





## Non-ideal factors

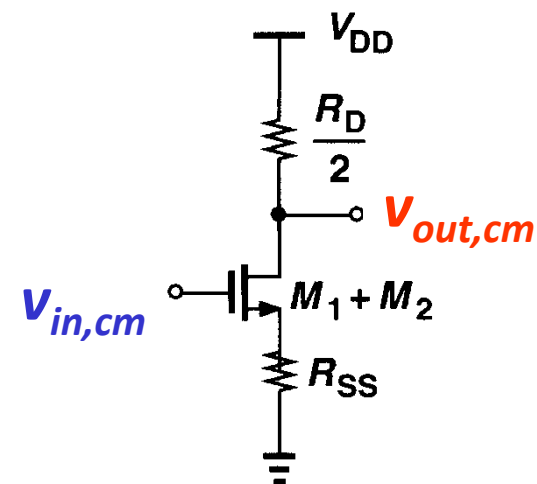
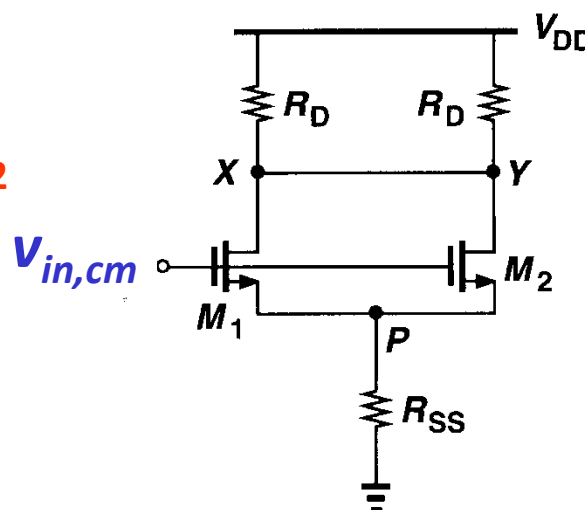
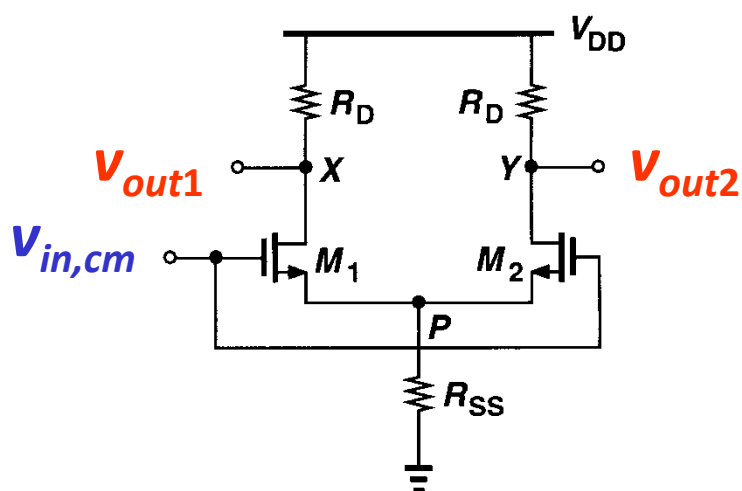
- Two non-ideal factors in diff. pairs:
  - finite output resistance of the tail current source
  - Mismatches





## Non-ideal tail current source

- Tail current source show finite output resistance
  - Consider the small-signal response of CM signal
  - Because of symmetry,  $V_{out1} = V_{out2} = V_{out,cm}$
  - *Simplified to a CM stage with source degeneration*

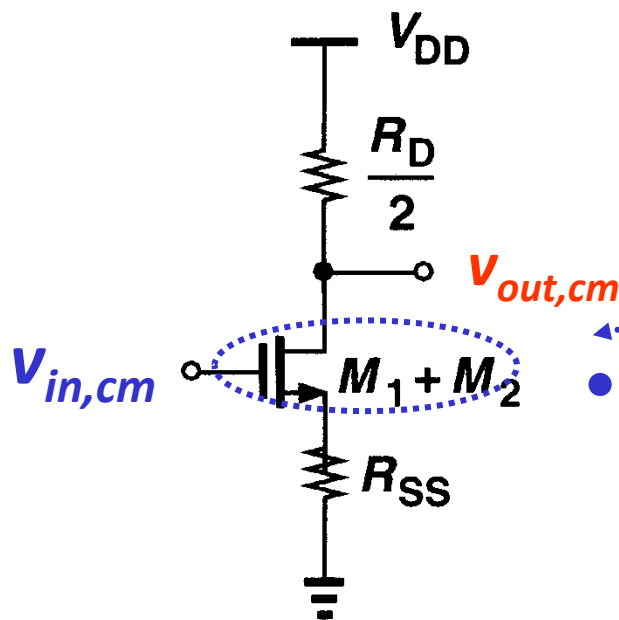


## CM small-signal gain



- The gain for the amplification of fluctuation in the input CM level

$$A_{v,CM} = \frac{V_{out,cm}}{V_{in,cm}} = \frac{R_D / 2}{1 / (2g_m) + R_{SS}}$$



- What effect is caused by finite  $A_{v,CM}$ ?
  - Instable biasing point
  - Output range is reduced (Ex. 4.6)
- The CM gain can be reduced with larger  $R_{SS}$ 
  - Increasing  $L$  of the tail transistor



# Influences of Mismatches: *CM to DM*



- #1: mismatches in the two resistors

- $V_{in,CM}$  increases by  $\Delta V_{in,CM}$

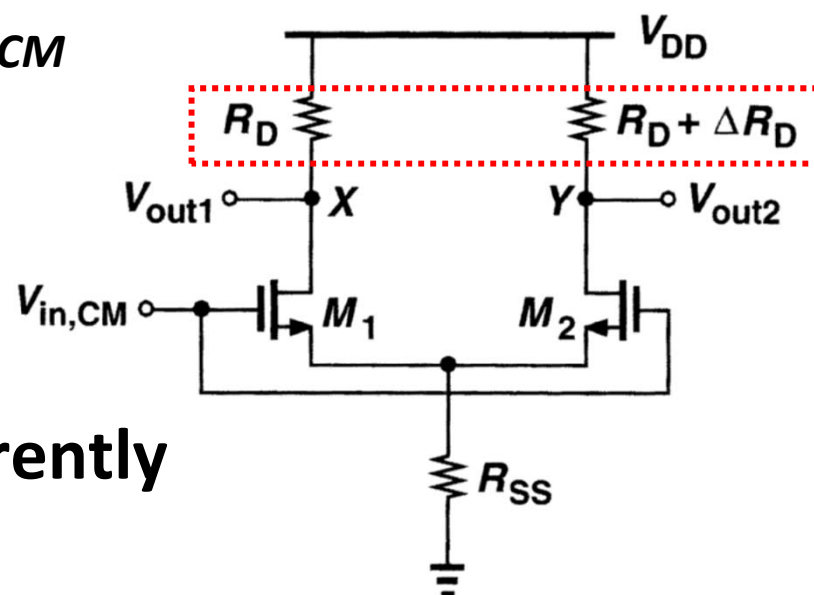
- $I_{d1}$  and  $I_{d2}$  increase by

$$\frac{g_m}{1 + 2g_m R_{SS}} \cdot \Delta V_{in,CM}$$

- $V_X$  and  $V_Y$  changes differently

$$\Delta V_X = -\frac{g_m}{1 + 2g_m R_{SS}} \cdot \Delta V_{in,CM} \cdot R_D$$

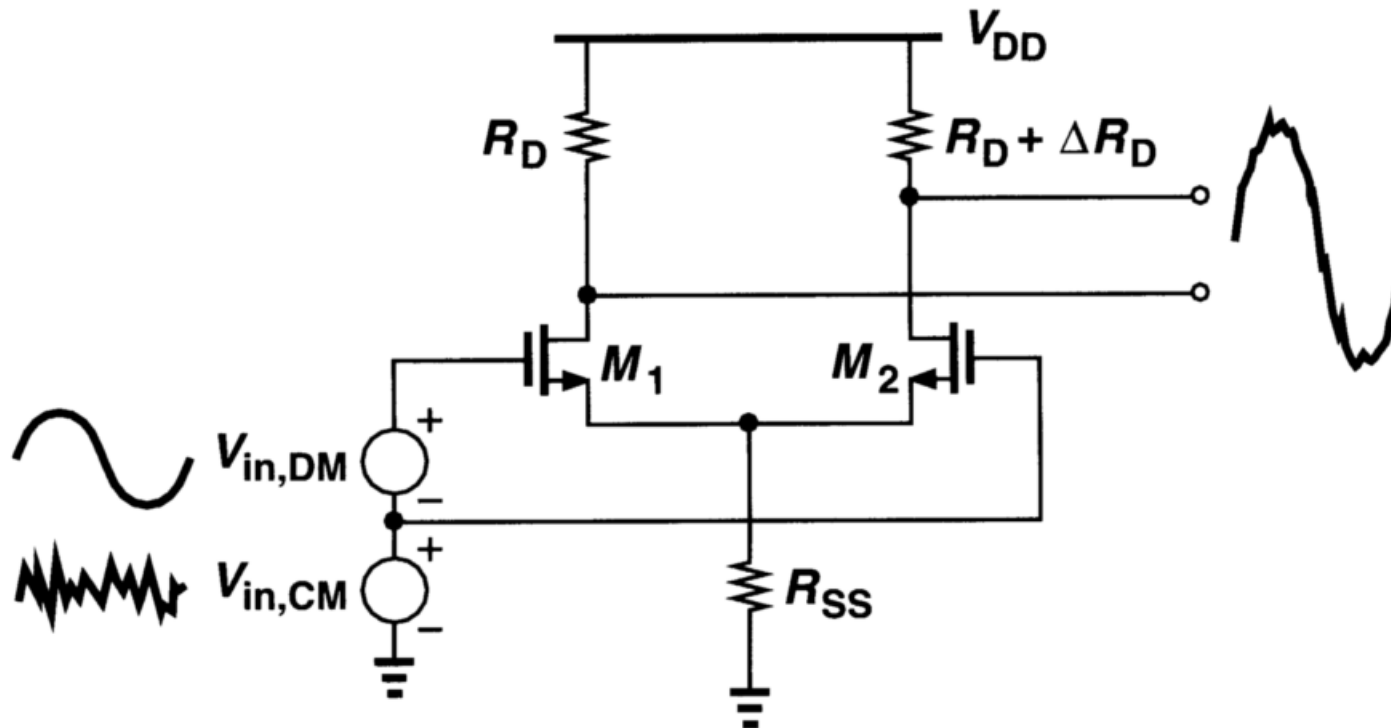
$$\Delta V_Y = -\frac{g_m}{1 + 2g_m R_{SS}} \cdot \Delta V_{in,CM} \cdot (R_D + \Delta R_D)$$



Variation in  $V_{in,CM}$   
converted to  
differential output!



# Effect of mismatches and finite $R_{SS}$



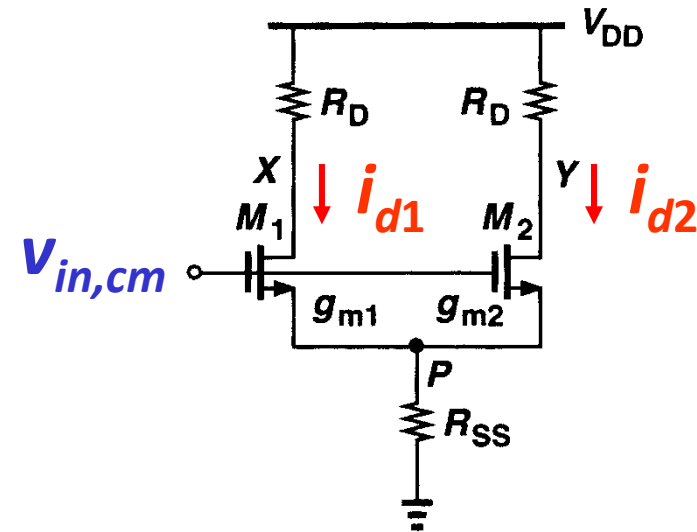
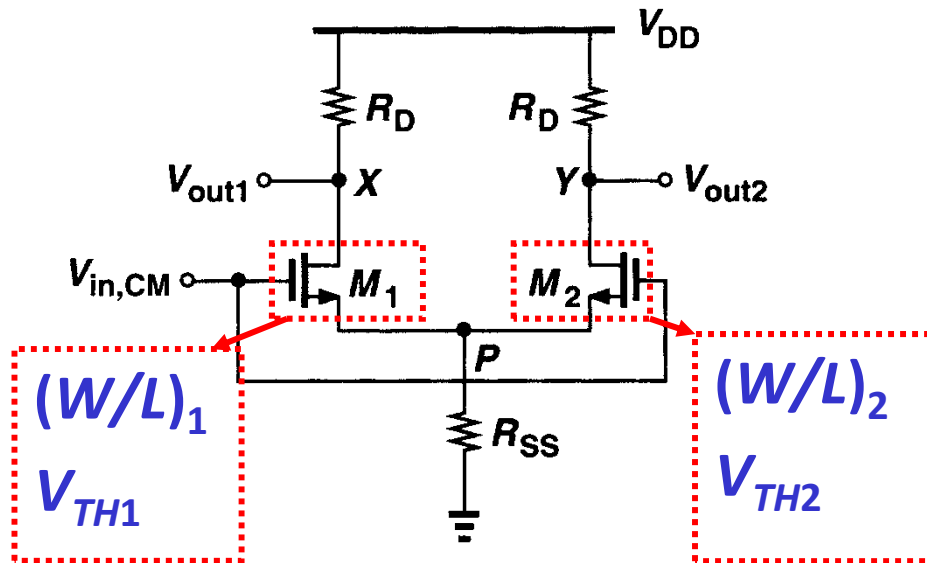
The differential output signal is corrupted by the common-mode input noise because of mismatches and the non-ideal tail current source



# Influences of Mismatches: $\Delta g_m$



- #2: mismatches in the two input transistors

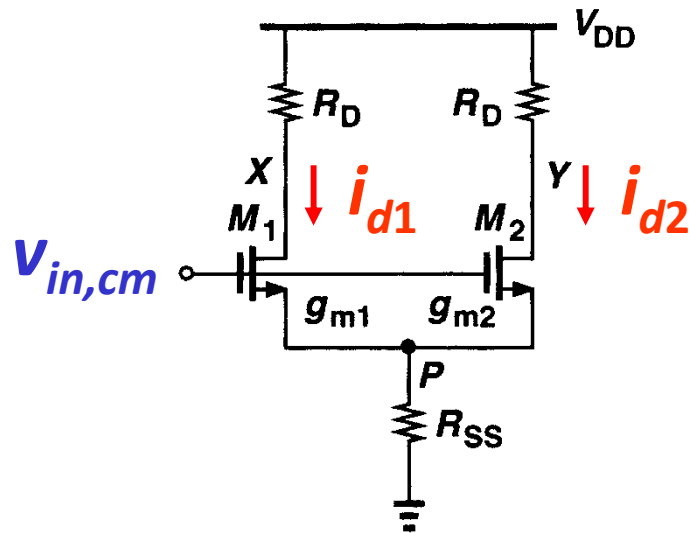


- The total small-signal current is  $i_{d1} + i_{d2}$

$$i_{d1} + i_{d2} = \frac{g_{m1} + g_{m2}}{1 + (g_{m1} + g_{m2})R_{SS}} \cdot v_{in,cm}$$



# Derivation of $A_{CM-DM}$



$$i_{d1} = \frac{g_{m1}}{1 + (g_{m1} + g_{m2})R_{SS}} \cdot v_{in,cm}$$

$$i_{d2} = \frac{g_{m2}}{1 + (g_{m1} + g_{m2})R_{SS}} \cdot v_{in,cm}$$

$$v_x = -i_{d1}R_D \quad v_y = -i_{d2}R_D$$

$$v_{out,diff} = v_x - v_y = -(i_{d1} - i_{d2})R_D = \frac{(g_{m1} - g_{m2})R_D}{1 + (g_{m1} + g_{m2})R_{SS}} \cdot v_{in,cm}$$

$$A_{CM-DM} = \frac{v_{out,diff}}{v_{in,cm}} = \frac{(g_{m1} - g_{m2})R_D}{1 + (g_{m1} + g_{m2})R_{SS}}$$



# Common Mode Rejection Ratio (CMRR)



- **CMRR:** Undesired CM-DM gain is normalized to the desired differential gain

$$g_m = (g_{m1} + g_{m2}) / 2 \quad \Delta g_m = (g_{m1} - g_{m2})$$

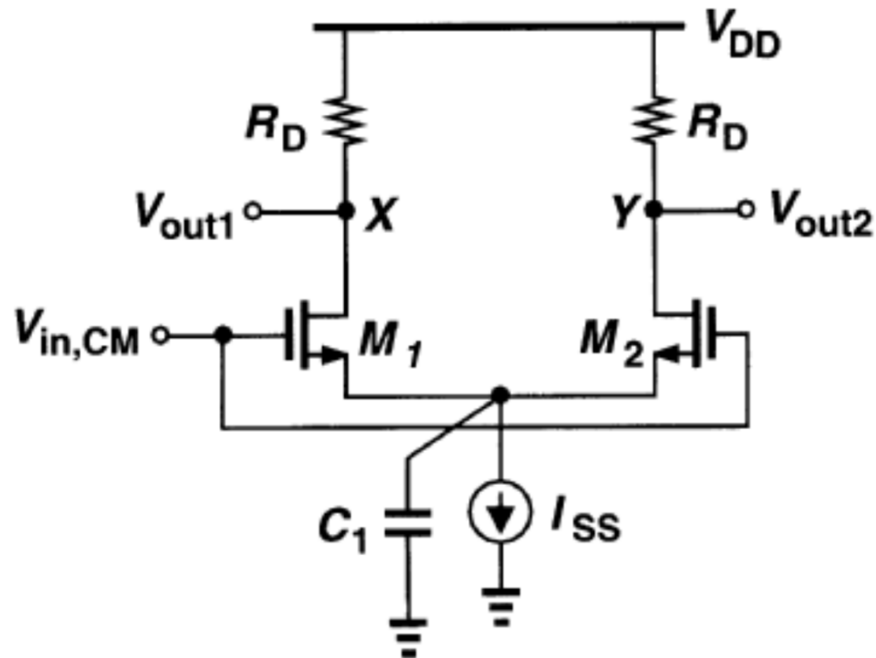
$$CMRR = \left| \frac{A_{DM}}{A_{CM-DM}} \right| = \left| \frac{g_m R_D}{\frac{\Delta g_m R_D}{(1 + 2g_m R_{SS})}} \right| \bullet \text{ To improve CMRR}$$

$$= \frac{g_m}{\Delta g_m} (1 + 2g_m R_{SS})$$

- Increase symmetry!
- Increase L of the tail transistor



# High-Frequency Common-Mode Response



- Importance of symmetry!

Parasitic capacitance reduces the output impedance of the tail current source !

# Outline

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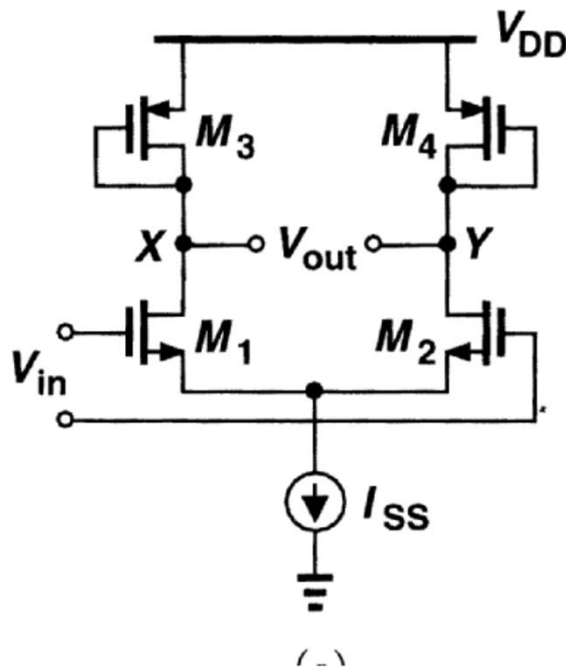


- Single-Ended vs. Differential
- Basic Differential Pair
- Common-Mode Response
- **Differential Pair with MOS Loads**
- Gilbert cell

# diode connected MOS Loads



- Linear resistors are rarely used as amplifier loads
- Diode-connected MOS loads
- Using half-circuit method:



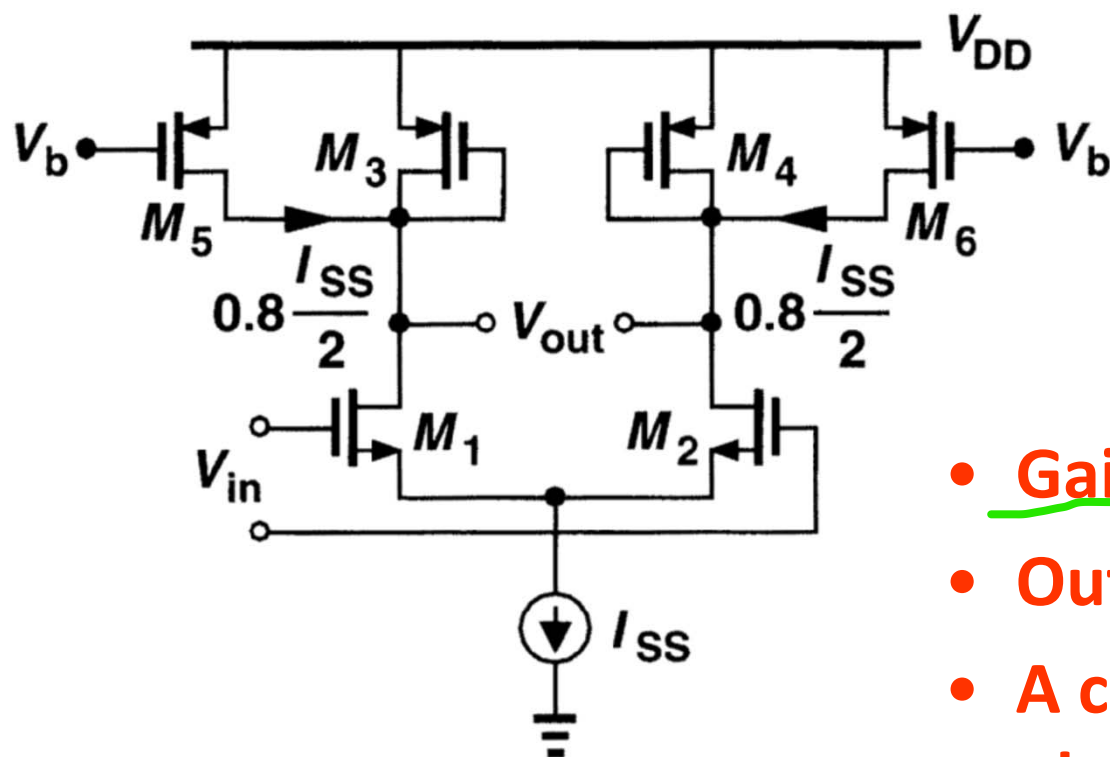
$$A_v = -g_{mN}(g_{mP}^{-1} \parallel r_{oN} \parallel r_{oP})$$
$$\approx -\frac{g_{mN}}{g_{mP}}$$



## Diode connected MOS Loads (cont.)

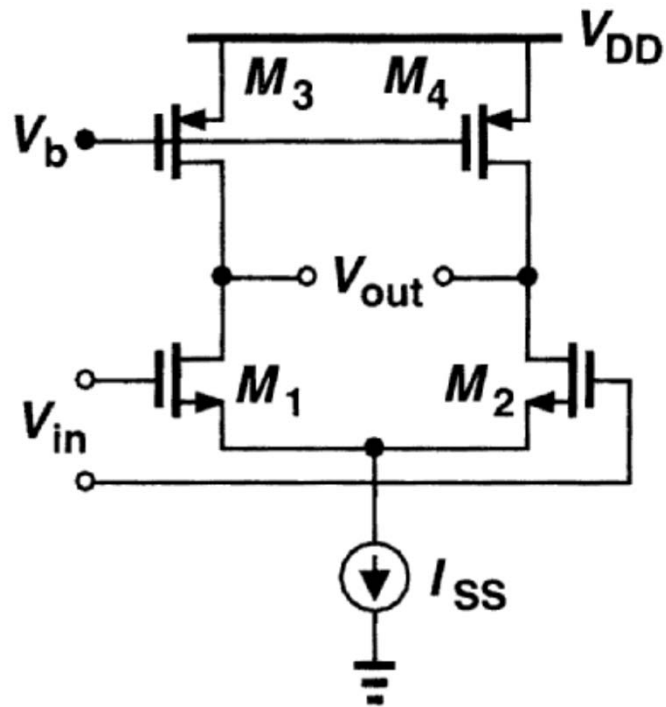


- Adding a parallel current source
  - A very useful circuit topology



- Gain is increased ✓
- Output range is enlarged
- A control terminal is obtained

# Current Source Loads



$$A_v = -g_{mN}(r_{oN} \parallel r_{oP})$$

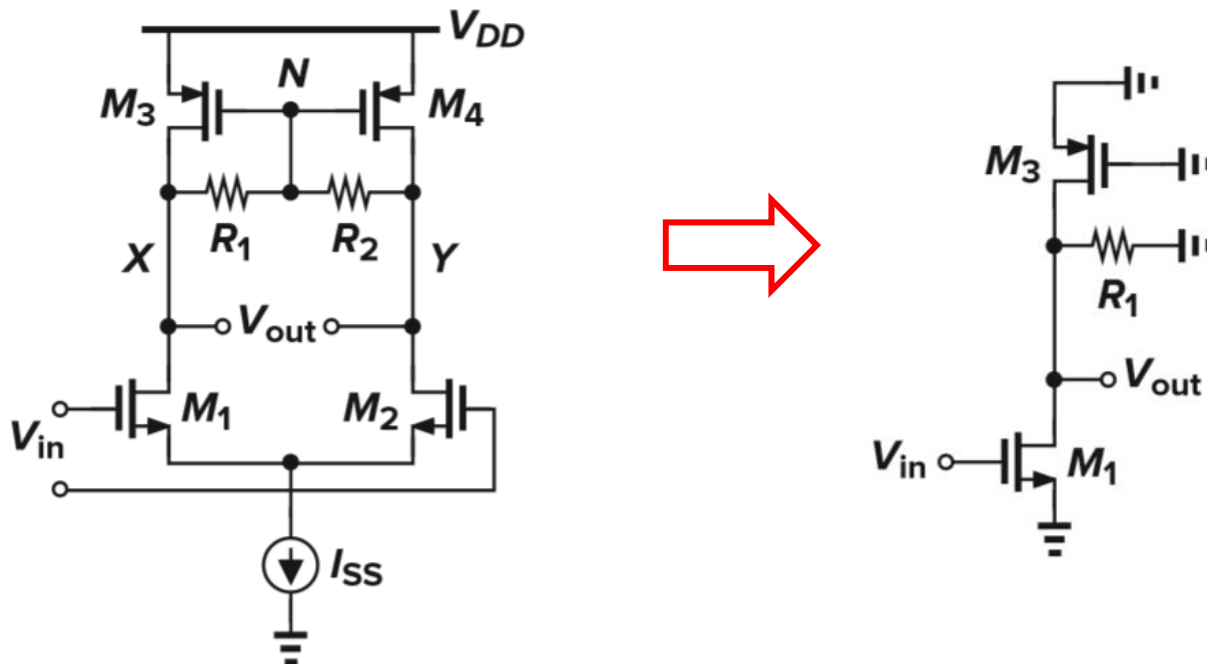


## Ex. 4.12



Simple and useful bias circuit:

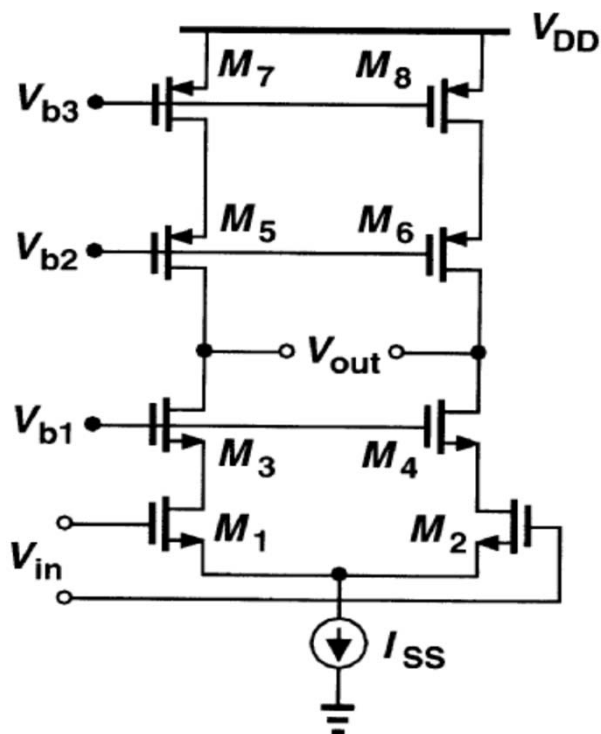
***N* is small-signal ground in diff. mode, why?**



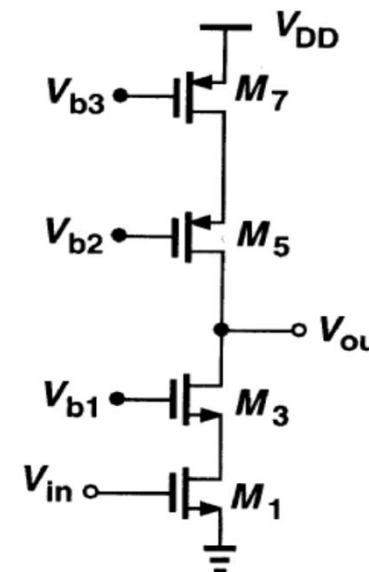
$$|A_v| = g_{m1}(r_{O1} \parallel R_1 \parallel r_{O3})$$



# Cascode Loads (cont.)



Half circuit



$$A_v \approx g_{m1}[(g_{m3}r_{o3}r_{o1}) \parallel (g_{m5}r_{o5}r_{o7})]$$

**Problem: the output common-mode level is not well defined!!**

*Will be solved in Chapter 9*



# Outline

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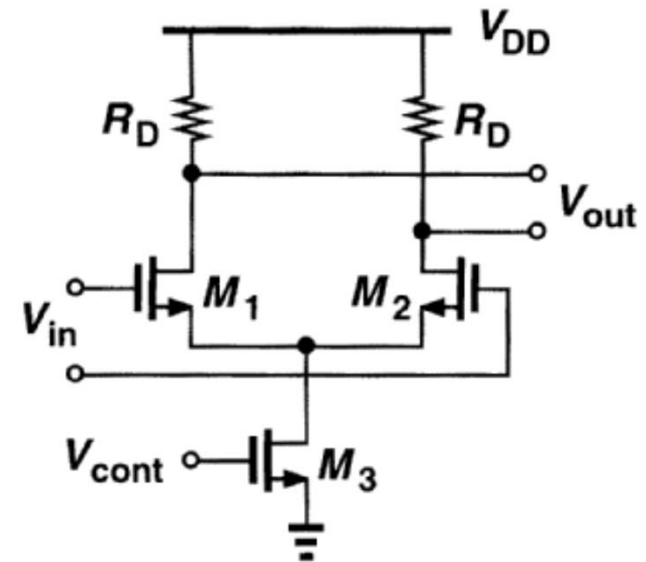
- Single-Ended vs. Differential
- Basic Differential Pair
- Common-Mode Response
- Differential Pair with MOS Loads
- **Gilbert cell**



# Variable Gain Amplifier (VGA)



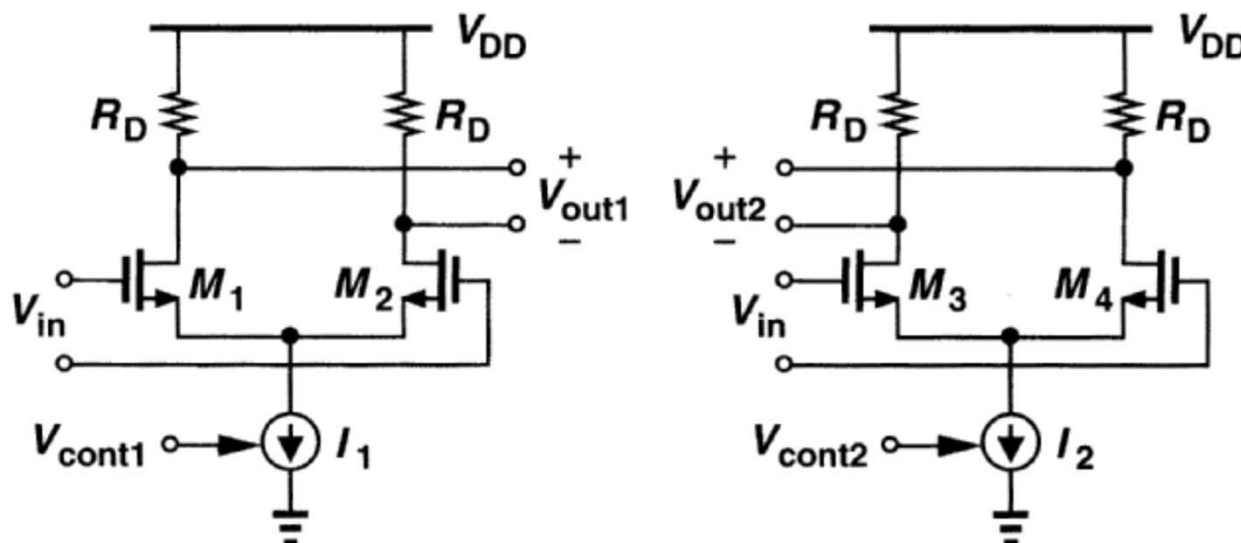
- In many applications, the gain of amplifiers needs to be adjustable.
  - Ex.: Mobile communication
  - A simple VGA
  - The gain is controlled by  $V_{cont}$ 
    - When  $I_{D3}=0$ ,  $A_v=0$ ;
    - $I_{D3} \uparrow$ ,  $|A_v| \rightarrow |A_v|_{max}$
    - $A_v$  is always negative, and  $|A_v|_{max}$  is limited by output range and device sizes





## Enlarge the gain range

- Can the gain vary from negative to positive?
  - Two amplifiers with opposite output phase
  - $v_{out1}/v_{in} = -g_m R_D$ ;  $v_{out2}/v_{in} = +g_m R_D$
  - Use two control signals

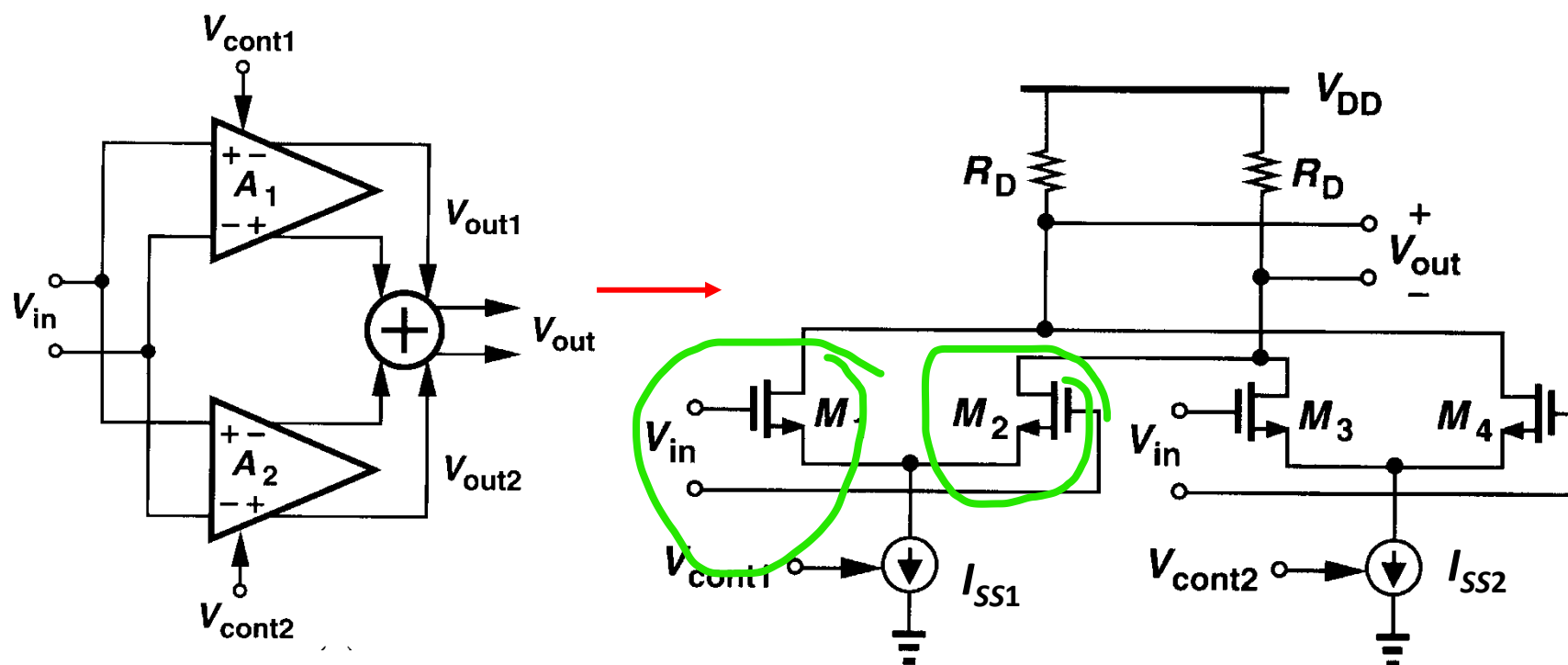


- How to combine into a single output?



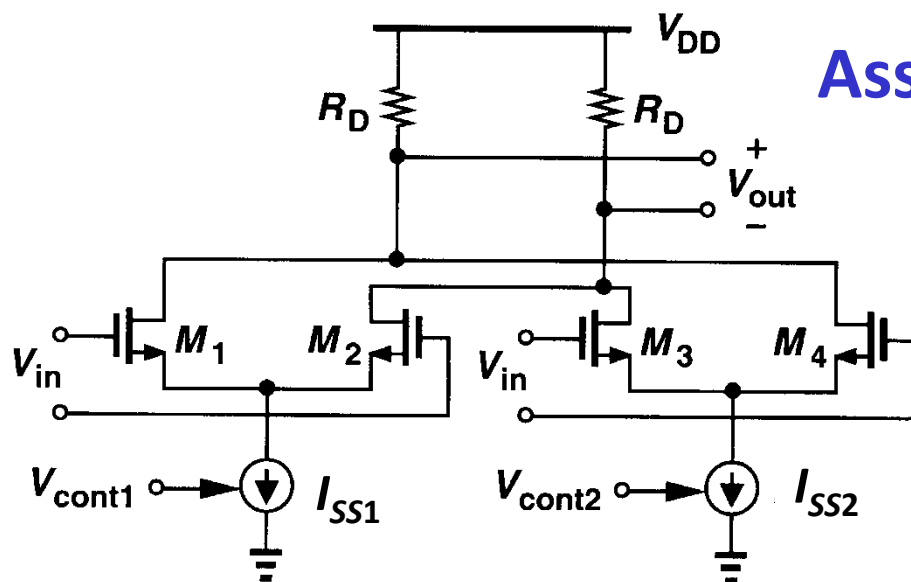
## Summation of the Two Outputs

- Difficult to realize voltage summation
- can be easily obtained in current domain



$$V_{out} = A_1 v_{in} - A_2 v_{in} \quad \checkmark$$





Assume  $\left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = \left(\frac{W}{L}\right)_3 = \left(\frac{W}{L}\right)_4$

– When  $I_{SS1} = 0$ ,

- $V_{out,diff} = +g_{m3,4}R_D$

– When  $I_{SS2} = 0$ ,

- $V_{out,diff} = -g_{m1,2}R_D$

$V_{out,diff} = -[(i_1 + i_4)R_D - (i_2 + i_3)R_D]$  – When  $I_{SS1} = I_{SS2}$ ,

$$= -[(i_1 - i_2) - (i_3 - i_4)] \cdot R_D$$

$$= -(g_{m1,2}v_{in} - g_{m3,4}v_{in}) \cdot R_D$$

- $V_{out,diff} = 0$

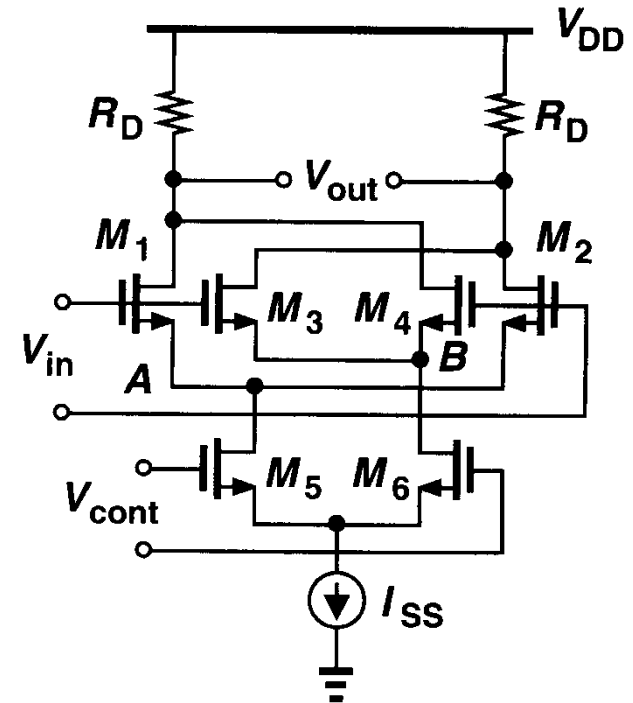
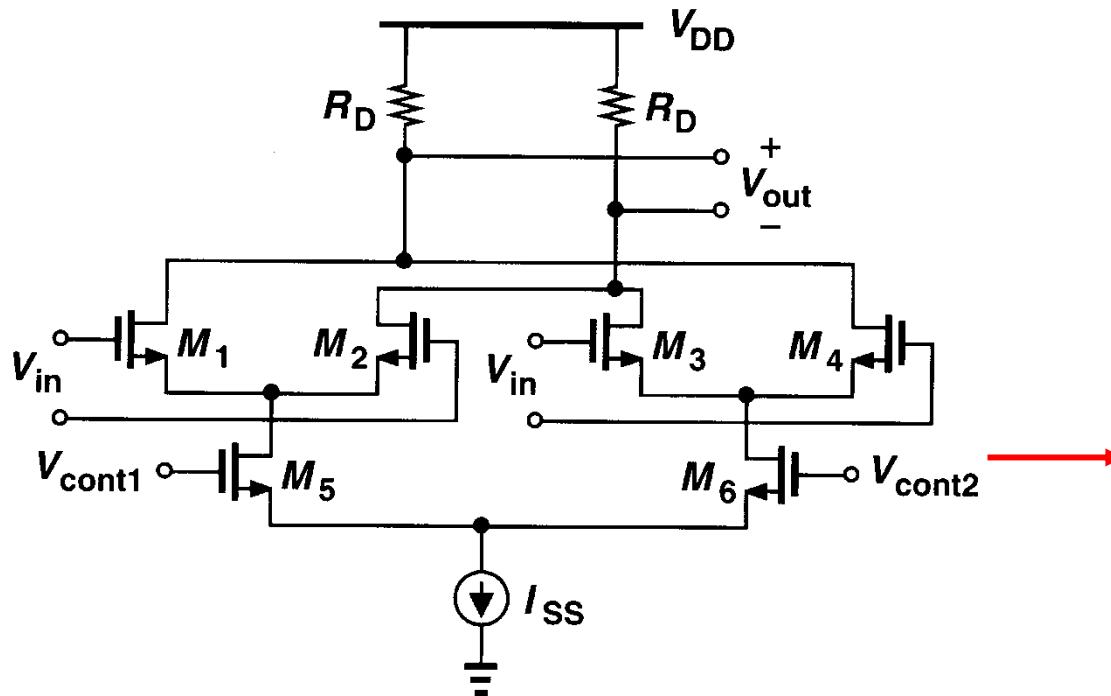


# Gilbert cell



- $V_{cont1}$  and  $V_{cont2}$  must varies  $I_{SS1}$  and  $I_{SS2}$  in opposite direction, how to realize?

– Use another differential pair → **Gilbert Cell**



**The input and output range are reduced by one  $V_{OD}$ !**



# Summary



- Single-Ended vs. Differential
  - Pros and cons?
- Basic Differential Pair
  - Large signal: valid input range
  - Small signal: half circuit concept
- Common-Mode Response
  - $A_{v,CM}$ ,  $A_{CM-DM}$ , CMRR
- Differential Pair with MOS Loads
  - Diode-connect, current source, cascode
- Gilbert cell